



Radiology imaging for GI TRACT Diseases

Bachtiar Murtala

Dept of Radiology Medical Faculty Universitas Hasanuddin

Modalities



X-Ray Abdomen
polos
(plain radiography)



X-ray + contrast
(barium sulphate,
Iodium)



Radiology Imaging
(USG, CT-Scan, MRI,
Nuclear)

Plain Abdominal Radiography

Very useful
particularly for acute
abdomen such as :
ileus, free air, free fluid

Stone, mass, etc

X-ray with contrast

Barium Sulfat, or
Iodium

To visualise
esofagus, gaster-
duodenum, small
bowel, large bowel

**Imaging
modalities
(Cross sectional)**

(hollow organs,
hepatobiliary,
Pancreas, Spleen,
peritoneal cavity)



USG



CT-Scan



MRI



Nuklir

Indication (most common)

Abdominal
pain/kolik

Tanda2 ileus

Free
air/fluid

Trauma
abdomen

Dispepsia

dysphagia

Abnormal
defecation

mass

infections

Congenithal
anomaly

Check-up
(mostly US)

diagnostic imaging approach for patients with acute abdominal pain. *RUQ* right upper quadrant, *RLQ* right lower quadrant, *PAR* plain abdominal radiography, *IVU* intravenous urography

PAR	US	CT	Comments
–	+	(+)	US preferred initial modality in acute cholecystitis, CT for evaluation of complications of acute cholecystitis
–	+	(+)	US for “screening,” CT when US findings equivocal
–	+	+	US preferred in children and in child-bearing women, CT initial modality in other patients with atypical clinical picture
–	(+)	+	CT preferred initial modality, US performed first when CT not available
(+)	(+)	+	Most conditions best visualized with CT; PAR may be initial modality in bowel obstruction
(+)	(+)	+	US or CT initial modality in suspected ureteral stones; PAR combined with IVU if US or CT not available; CT modality of choice in acute pancreatitis

– not helpful for diagnosis, (+) helpful adjunct but only in special situations, + useful for establishing diagnosis

in acute abdomen

3 standard positions

1. Supine
2. Erect (tegak)
3. LLD (Left Lateral Decubitus)

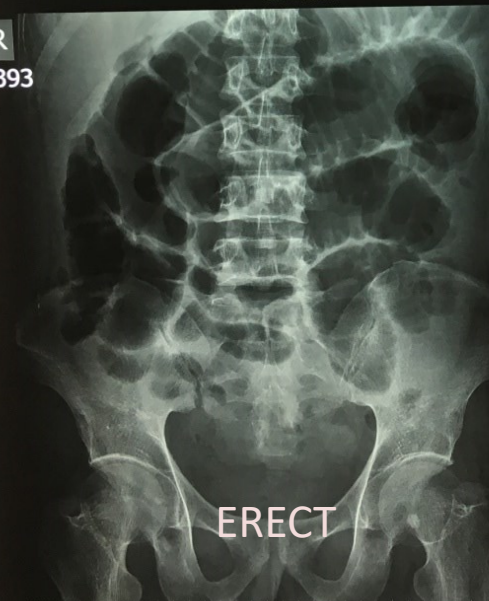
LLD
0393



02/02/2019

69.8 %

R
0393



ERECT

02/02/2019

54.3 %



RS Akademis Jaury Jusuf Putra
INSTALASI RADIOLOGI
Jl. Jend. M. Jusuf No.57A Makassar

R



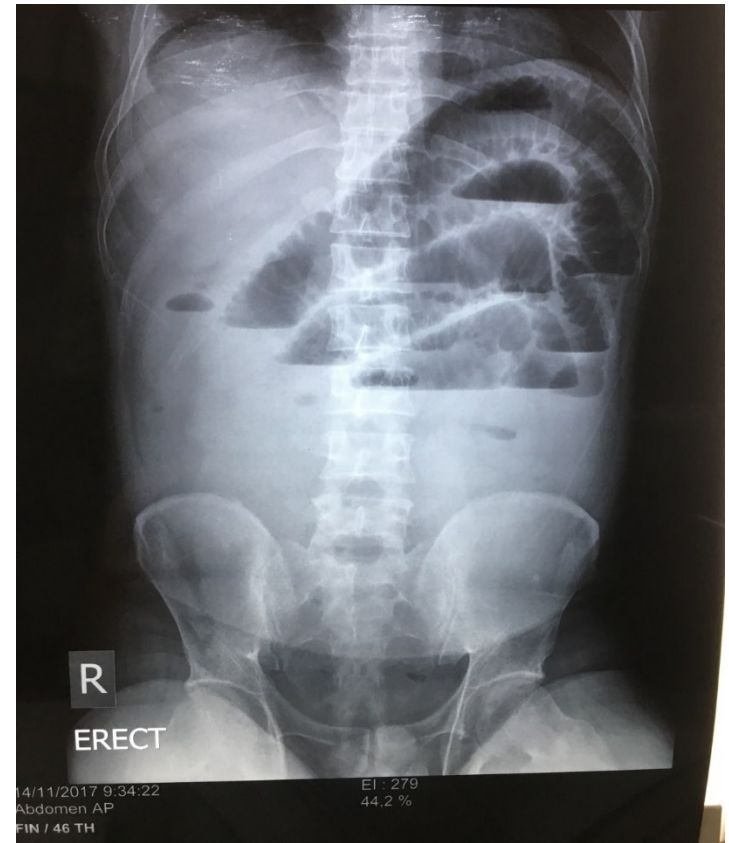
TEGAK

02/02/2019

54.2 %

Name : ARIADY DACHLAN, TN
Patient ID : 228942
Age : 40 year(s) Male
Date : 02/02/2019 11:43:06
Radiografer : Andhi

Pasien dewasa dgn perut kembung dan nyeri kolik,
muntah(+)



ILEUS OBSTRUKSI

Pasien dewasa dgn kolik abdomen



Frimann-Dahl sign



VectorStock

vectorstock.com/22731878



sigmoid volvulus: 'Coffee bean' sign.

Volvulus



Sigmoid volvulus - coffee bean sign

Sigmoid volvulus classically results in the formation of a loop of sigmoid colon, which is twisted at the root of the sigmoid mesentery, which lies in the left iliac fossa (LIF).

The loop of dilated bowel usually points upwards towards the diaphragm





pneumoperitoneum: note the Rigler's sign (arrowheads).



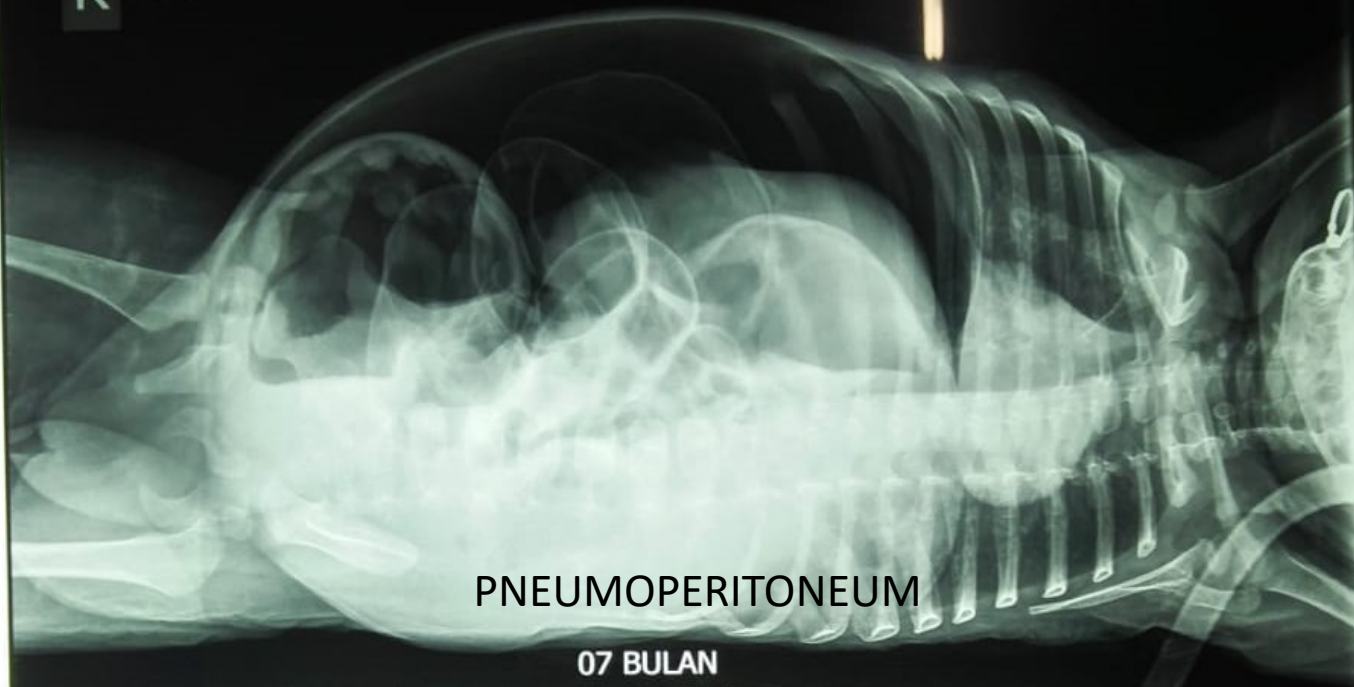
pneumoperitoneum: free air under both hemidiaphragms.



Pasien kembung.
Riw trauma(-)

PNEUMOPERITONEUM

R 2551



PNEUMOPERITONEUM

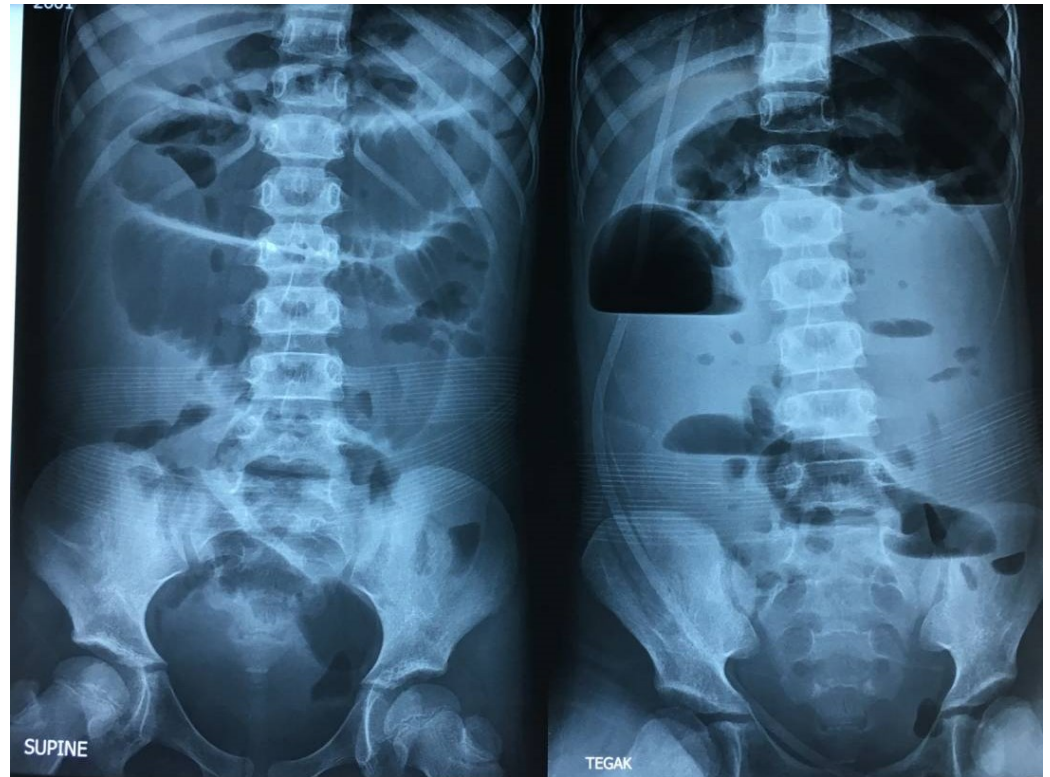
07 BULAN

24/07/2018

True size

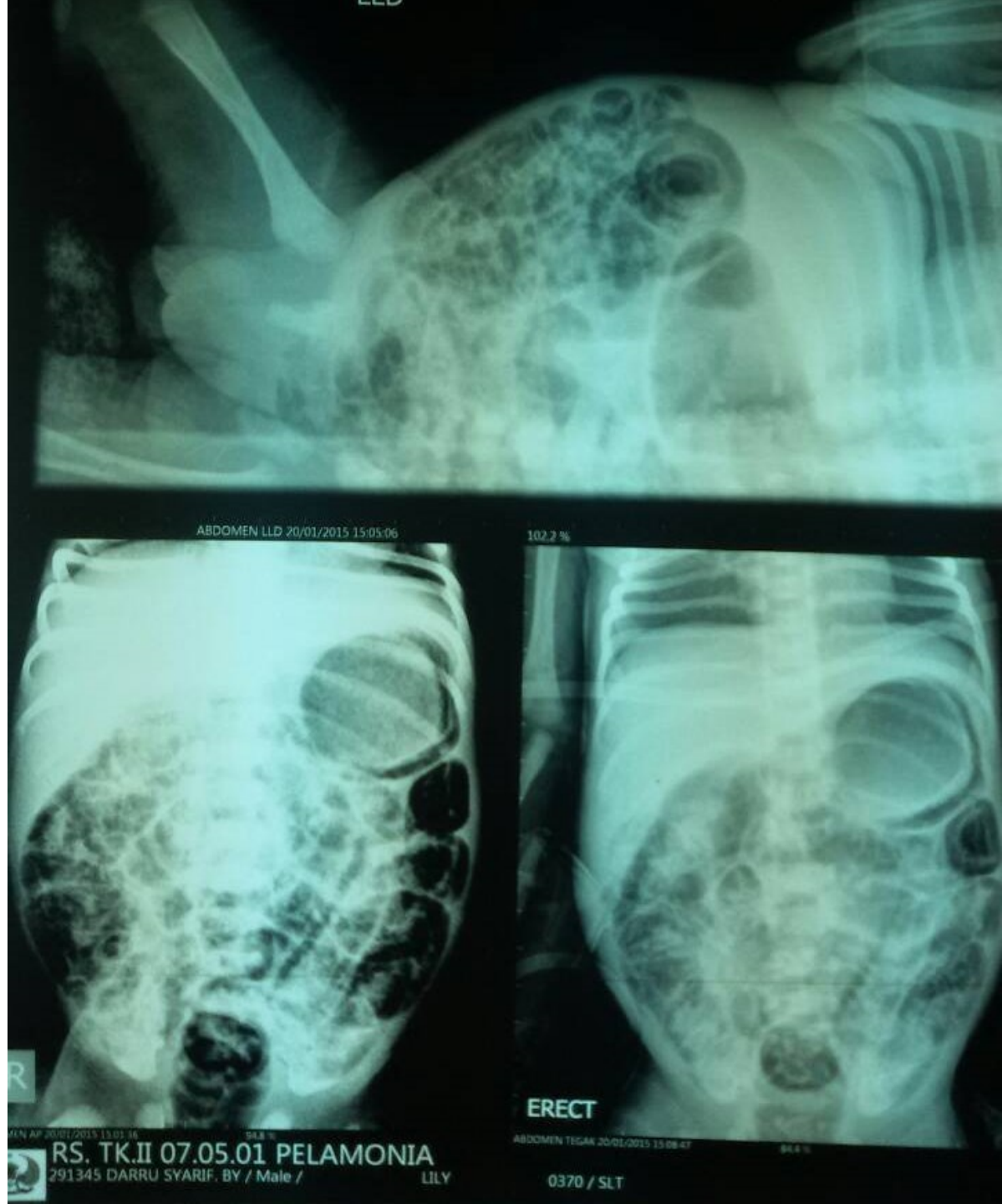


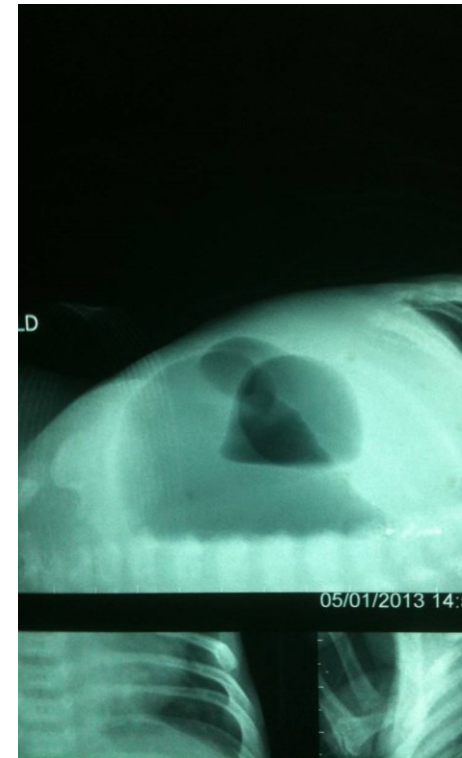
M, dgn
perut
membesar
(Ascites)



Pneumatosis intestinalis

Bayi dgn perut
kembung,
demam
(Necrotizing
Enterocolitis)





Bayi 2 bulan dengan riwayat muntah stlh menyusui

STENOSIS DUODENUM
(Double bubble sign)

Bayi 7 bulan dgn Bab campur darah dan lendir



Intussuseption = invaginasi (soft tissue mass, sirkuler)



Foto Dengan Kontras

- Mengevaluasi apakah aliran/passase kontras lancar atau tidak
 - Menemukan tanda2 obstruksi/stenosis
 - Mencari *filling defect* (cacat pengisian) atau *additional shadow* (bayangan tambahan). Dll
- 

Jenis pemeriksaan

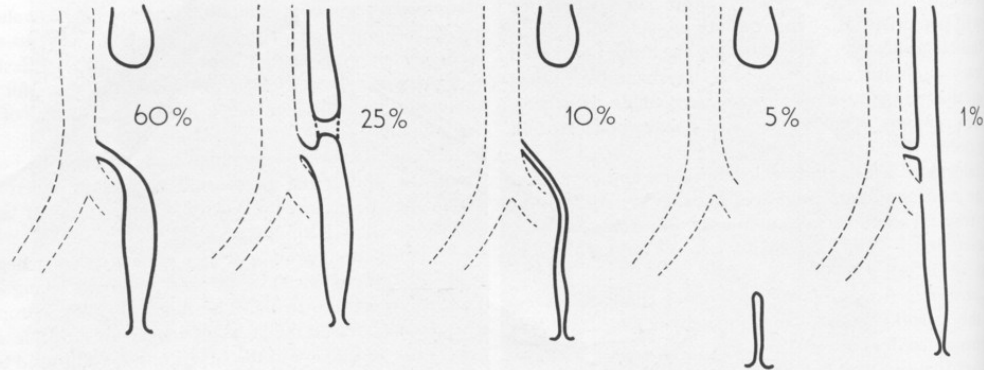
- Esofagus (**Esofagografi**)
- Gaster-Duodenum (**Gastroduodenografi** atau **MD foto**)
- Usus halus (**follow through**)
- Usus besar (**colon inloop** atau **Barium Enema**)

Bahan Kontras : **BARIUM SULFAT (BaSO₄)** atau **IODIUM water soluble**

Bayi baru lahir sll muntah saat menyusu

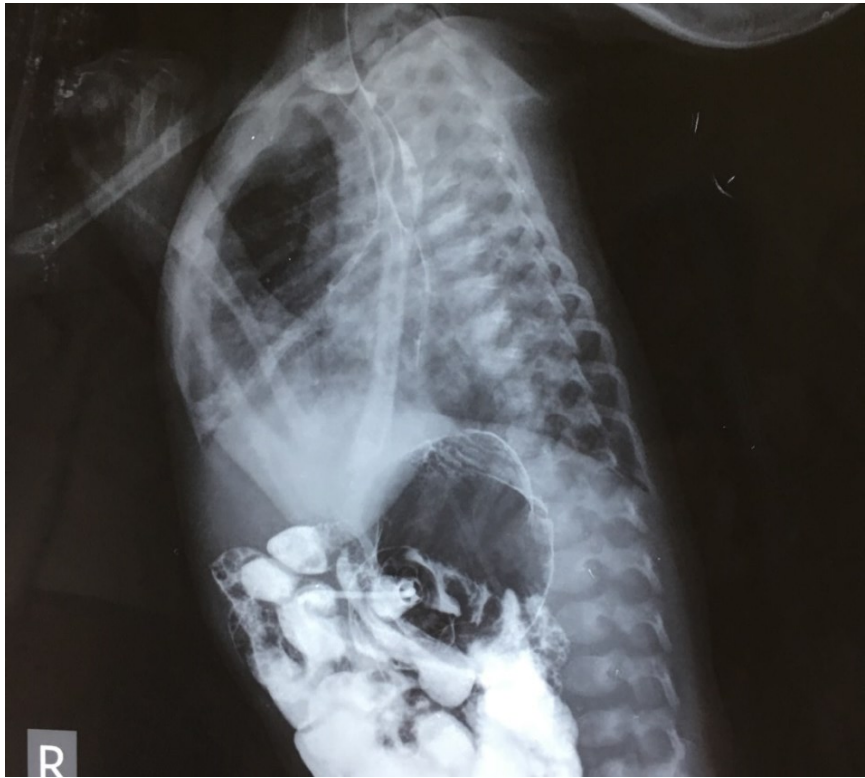


Esophageal atresia



Drawing showing the types of tracheo-oesophageal atresia. The relationship of the trachea to the lower segment of the oesophagus is important in determining the presence of air in the stomach.



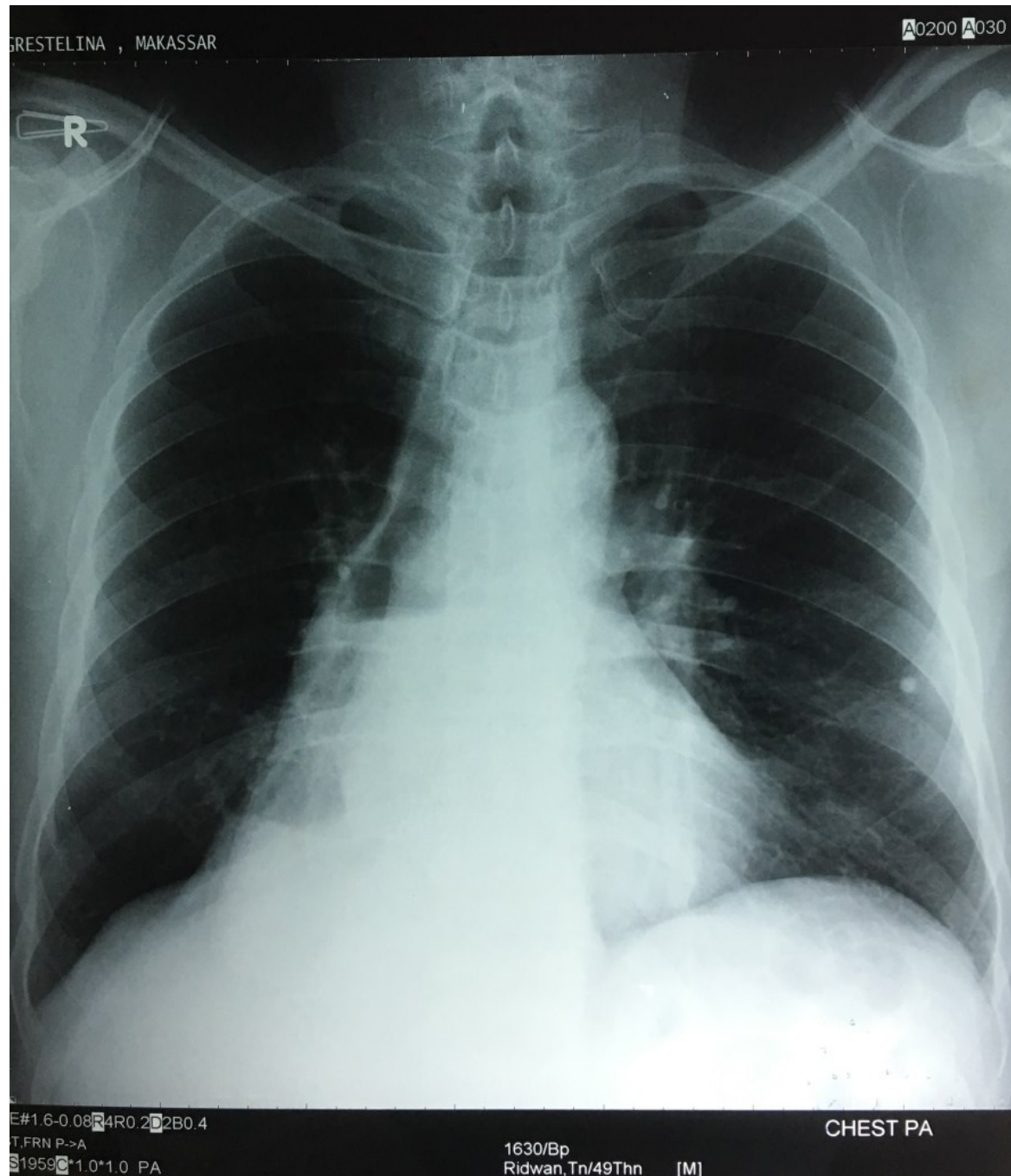


Bayi sering sesak (Hernia diafragmatika kiri)



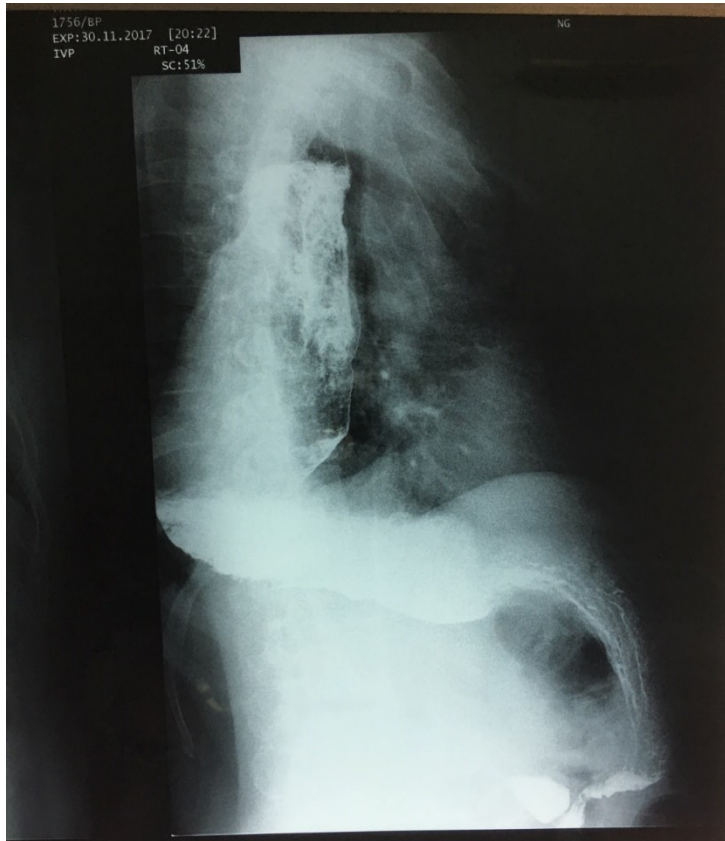
Laki2 dewasa
dgn gangguan
menelan

Achalasia



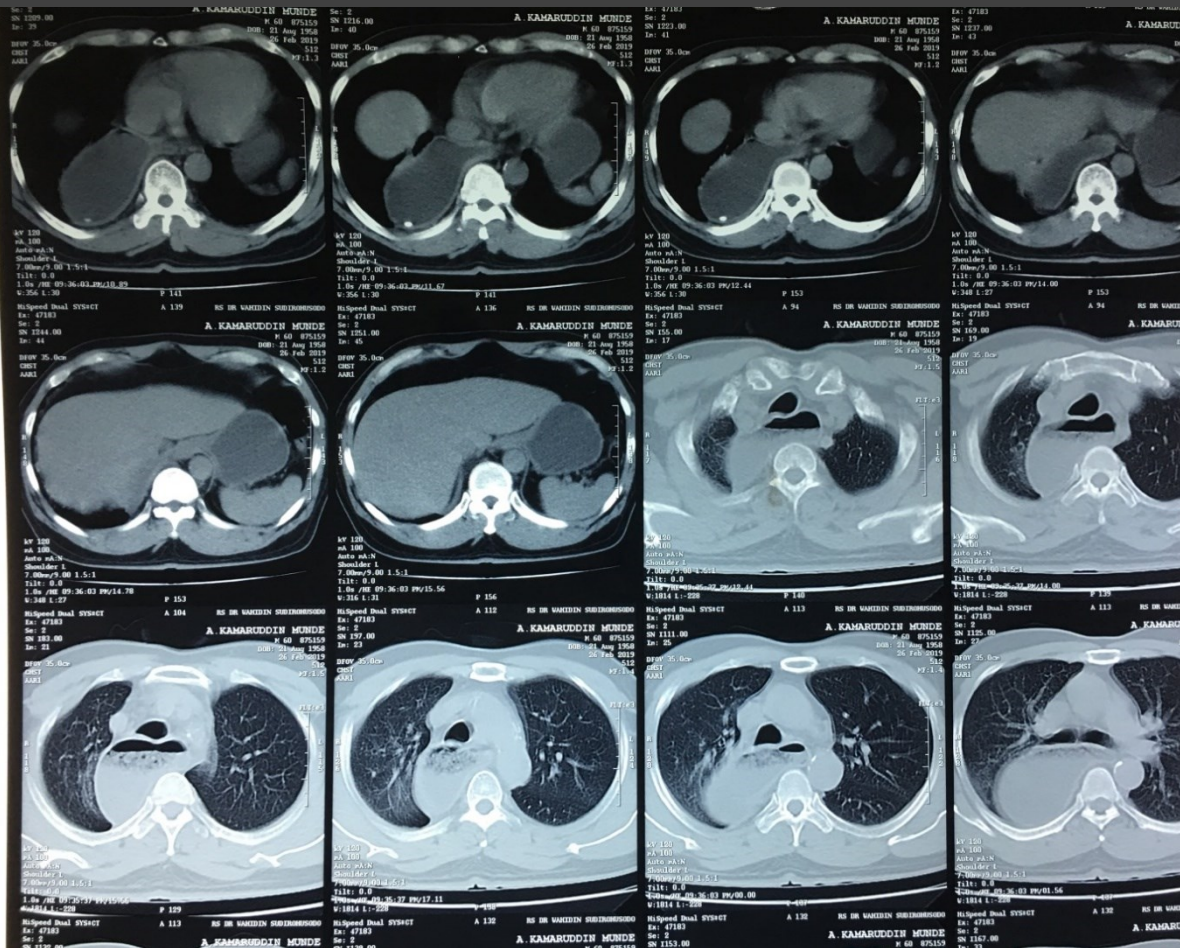


esofagografi

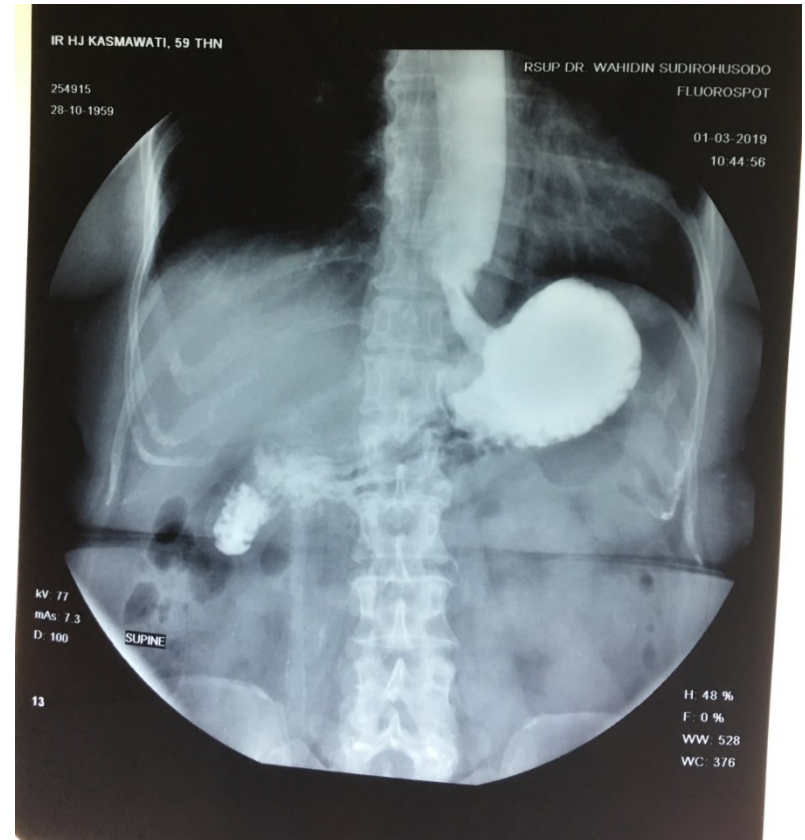
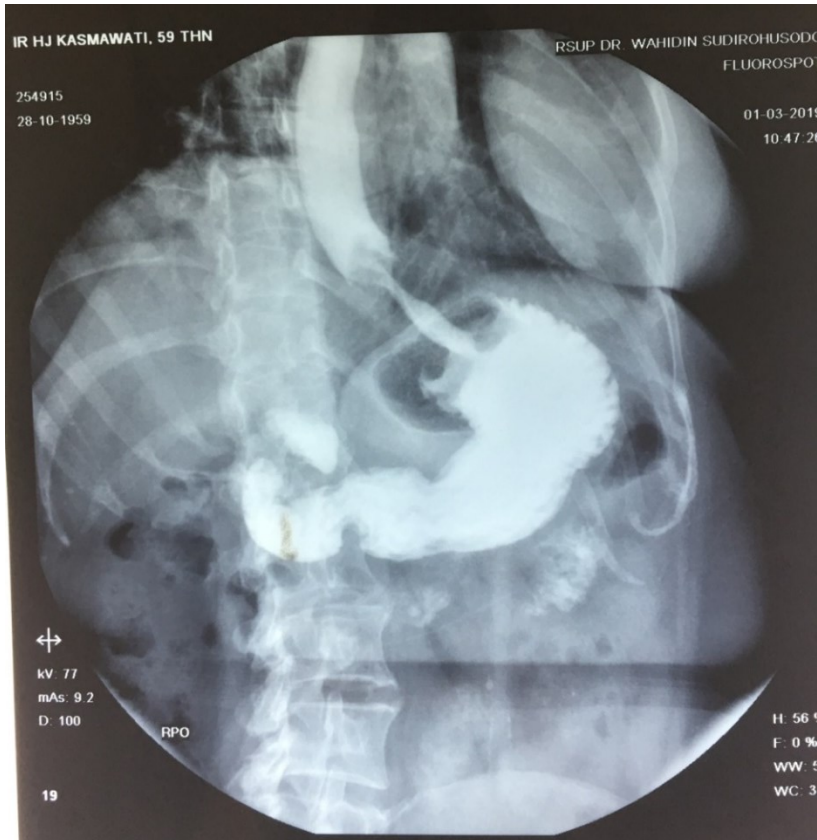


Mouse tail appearance, atau
Bird-beak appearance

CT scan



Tumor esofagogastrik (hematemesis melena) tampak filling defect irregular



Nyeri epigastrium

Ulkus gaster, tampak barium spot

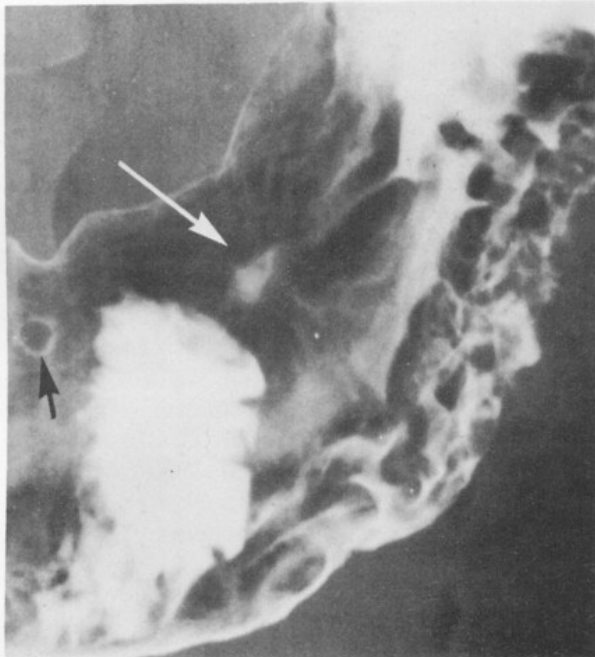


Fig. 33.21 Gastric ulcer seen in double contrast with the folds extending into the ulcer crater (*white arrow*). Black arrow shows the characteristic appearance of an air bubble. Note that the inner margin of the air bubble is sharply defined while the outer edge of the barium surrounding the air bubble is indistinct. This is the complete reverse of the appearances in acute gastric erosion.

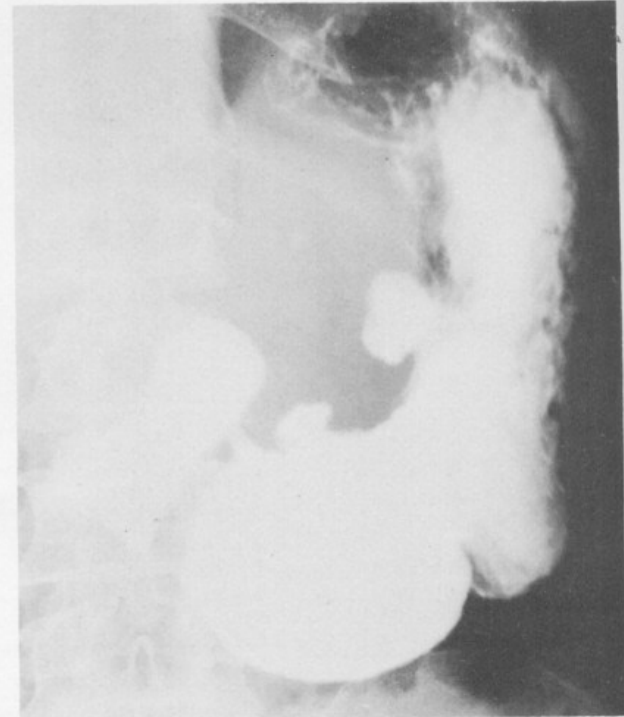
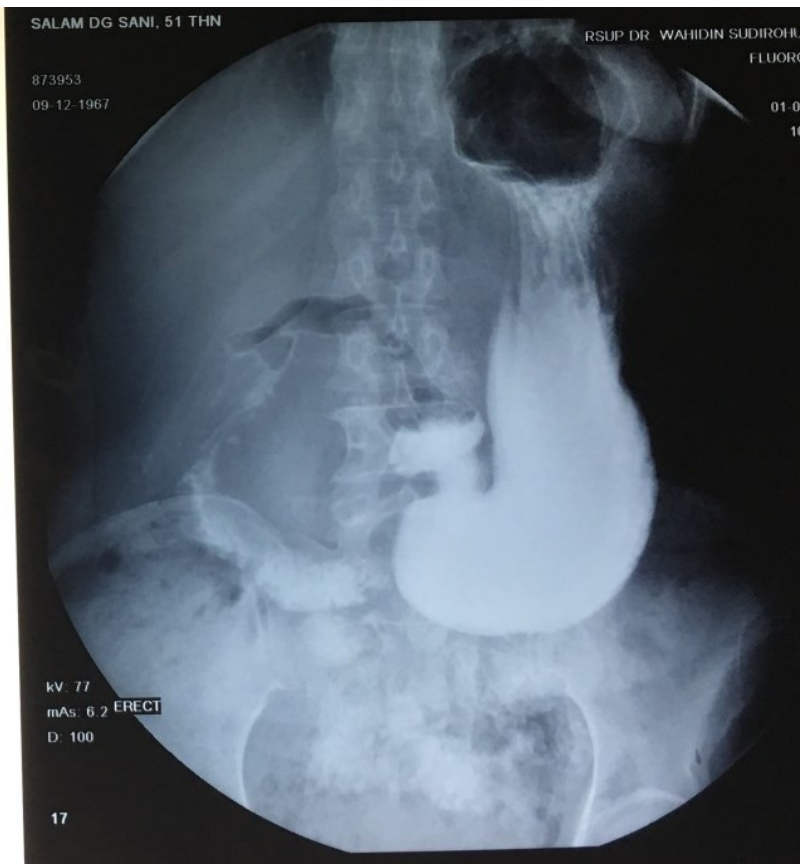
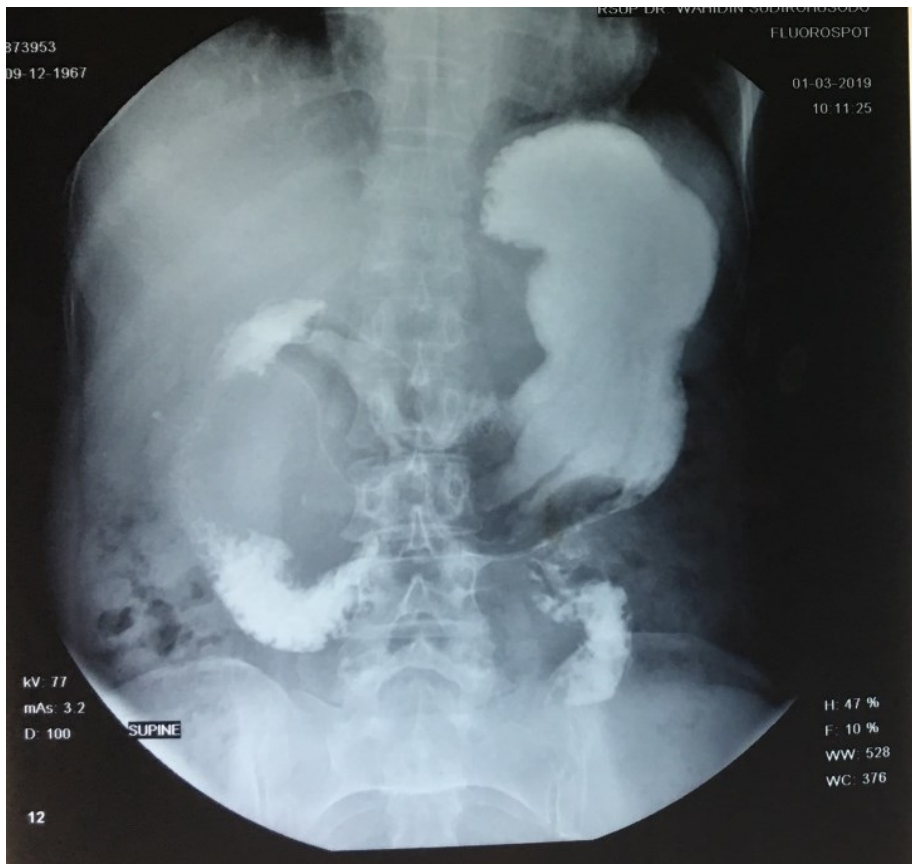


Fig. 33.18 Multiple gastric ulcers. A feature which is frequently noted in association with a delta-celled tumour of the pancreas (Zollinger-Ellison syndrome).

Teraba massa epigastrium (extralumen, kaput pankreas)



SALAM DG SANI, 51 THN

873953

09-12-1967

RSUP DR. WAHIDIN SUDIROHUSODO
FLUOROSPOT

01-03-2019
10.12.45



kV: 77
mAs: 4.3
D: 100 RPO

15

SALAM DG SANI, 51 THN

873953

09-12-1967

RSUP DR. WAHIDIN SUDIROHUSODO
FLUOROSPOT

01-03-2019
10.12.45

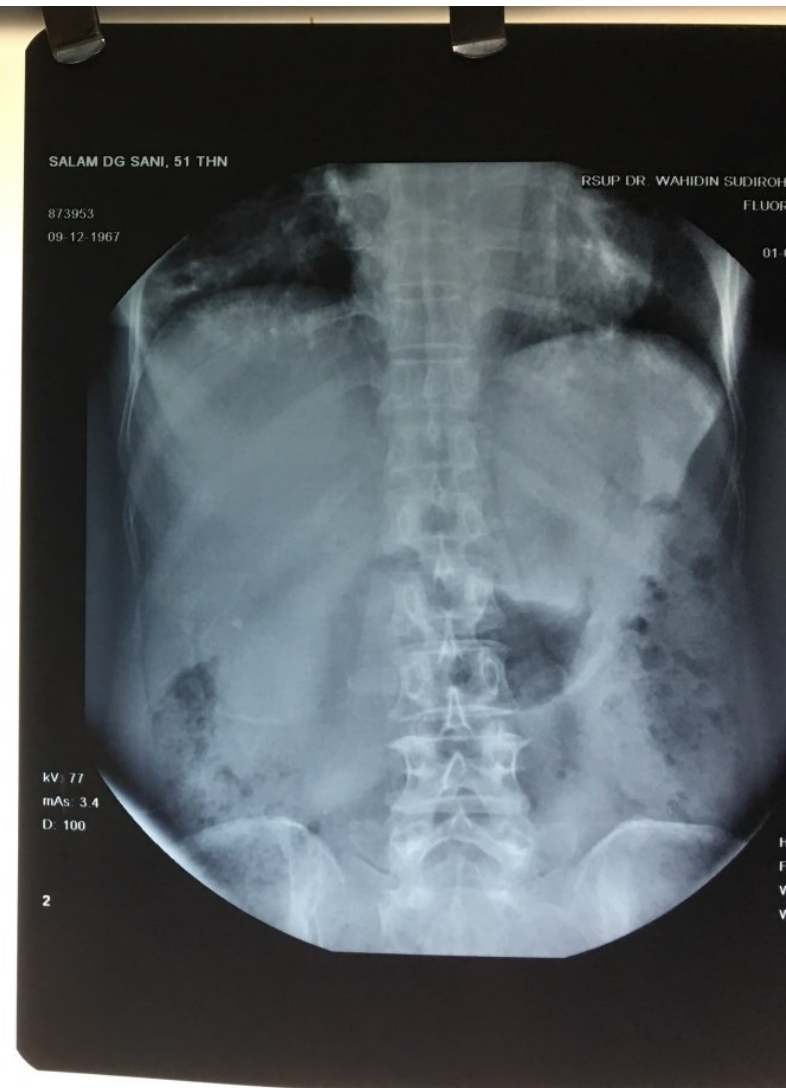
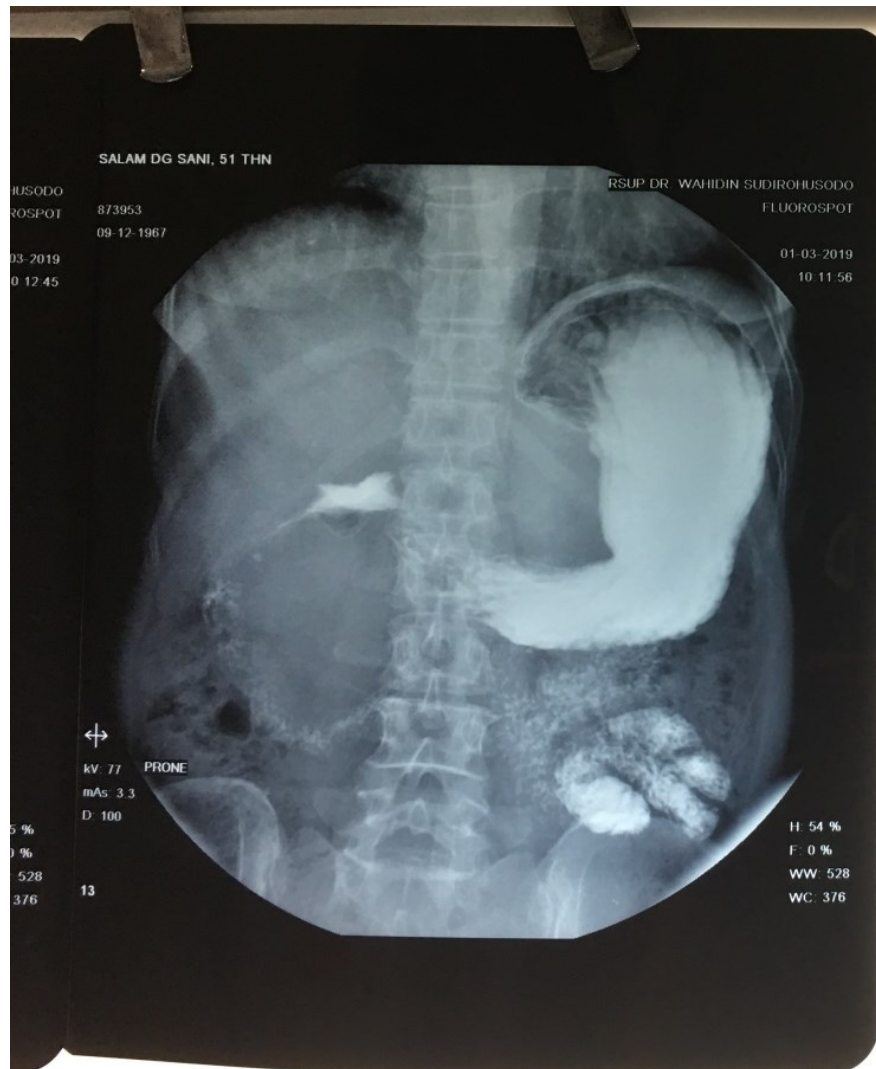


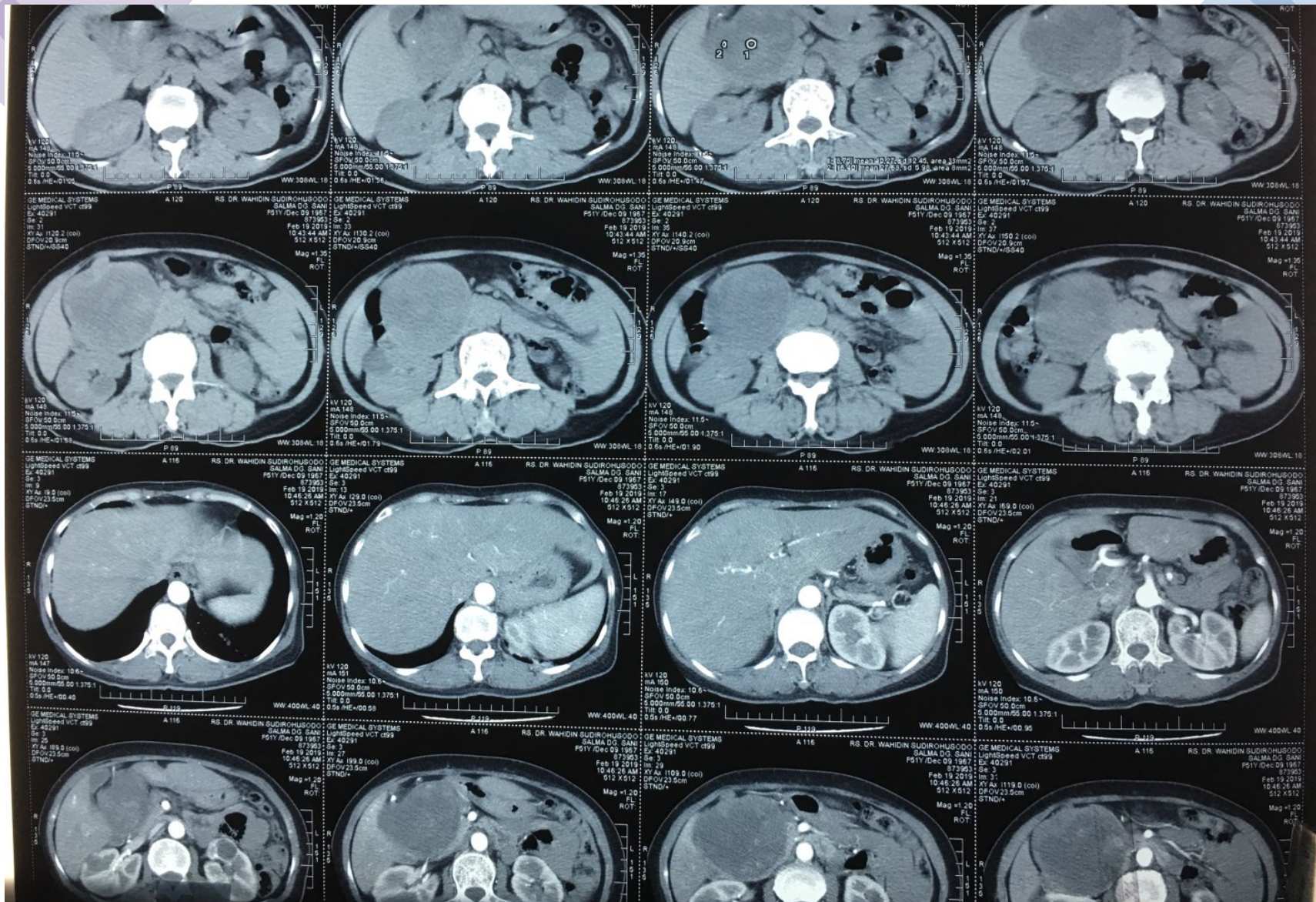
kV: 77
mAs: 3.3
D: 100
PRONE

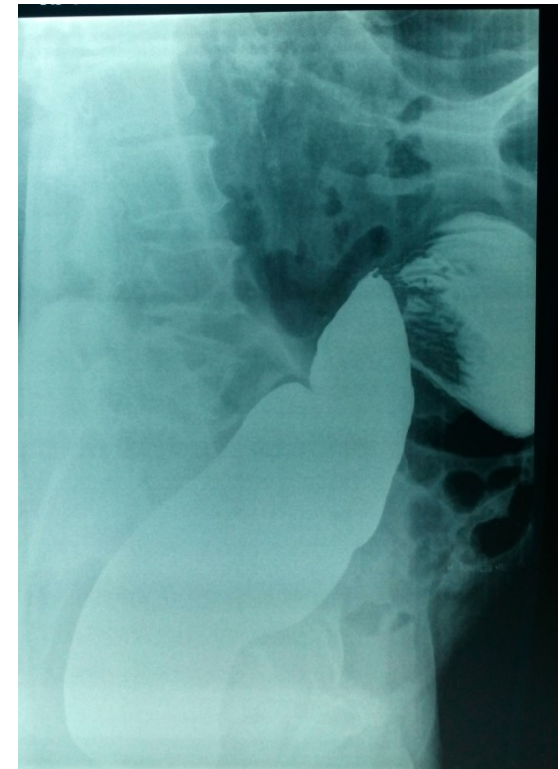
H: 45 %
F: 10 %
WW: 528
WC: 376

13

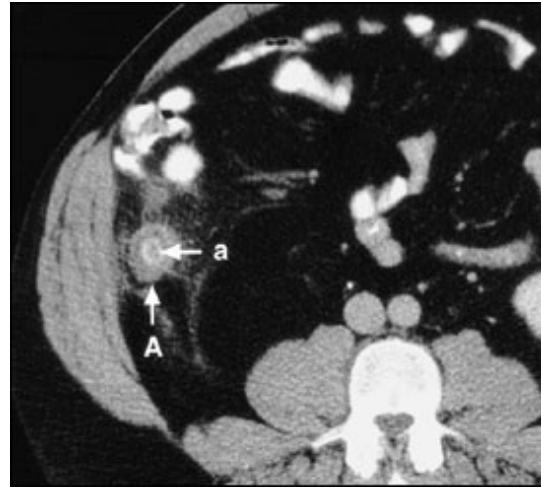
H: 54
F: 0.9
WW:
WC: 3







Sakit perut akut, kolik
Coffee bean, bird beak, whirlpool sign



Nyeri abdomen kanan bawah

Appendicitis acute

- Anak, sulit BAB
- Megacolon Congenital
- (Hirschsprung disease)



- Hematokhesia,
- Filling defect rectosigmoid
- akibat cancer



Inflamasi:

Ulcerative
colitis

Crohn's
Disease

- **Ulcerative Colitis**

- Loss of haustra
- Contracted, shortened & small calibre
- Saw-toothing/ulceration
- “Stringiness/String sign”

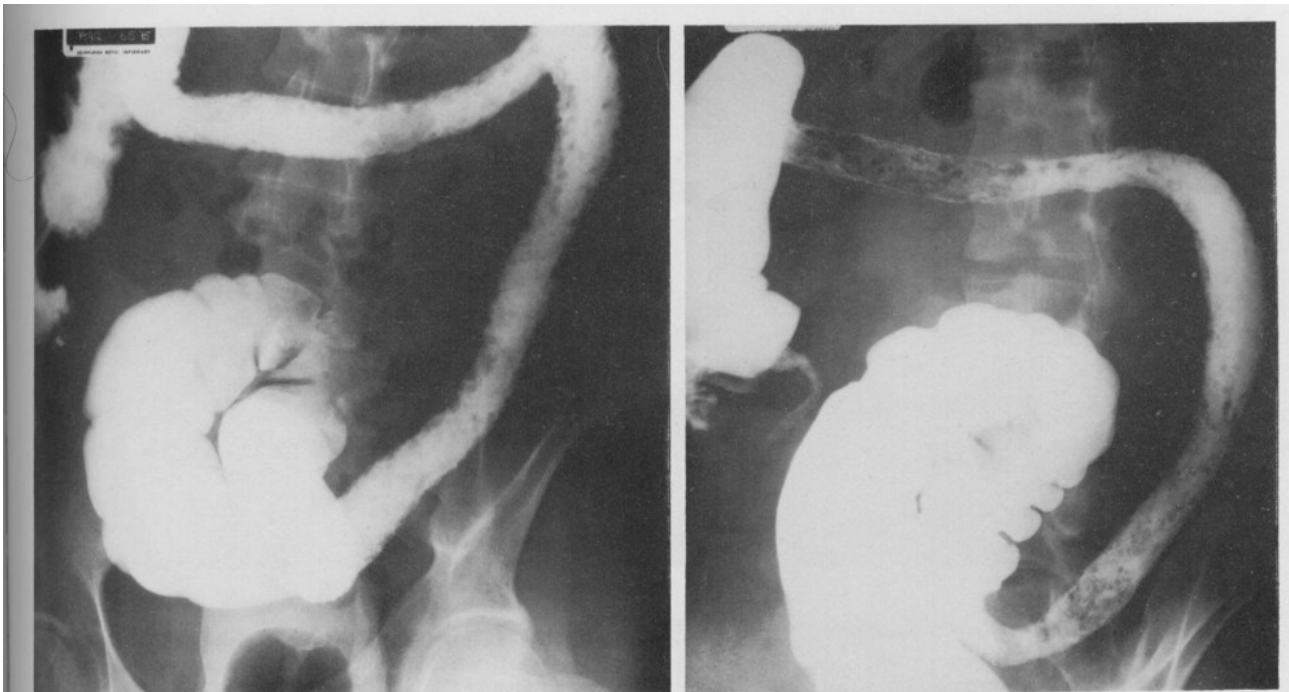
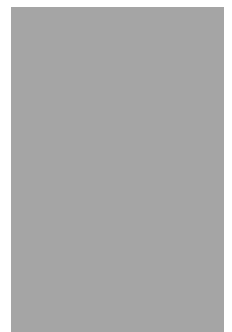


Table 2.1 Features of inflammatory bowel disease

Features	Crohn's disease	Ulcerative colitis
Distribution	Anywhere in gastrointestinal tract, mainly terminal ileum	Predominantly colon and rectum
Morphology	Skip lesions, aphthous ulcers, deep (rose thorn)	Disease involve continuous segments, superficial ulceration, aphthous ulcers (Fig. 2.27)
Fistula and sinuses	Common	Uncommon
Strictures	Yes	Relatively more common
Toxic dilatation	Uncommon	Common (Fig. 2.28)
Rectal involvement	Less than 50%	Greater than 95%
Anal fistula, sinus and fissures	Common	Uncommon
Terminal ileum	Narrowed, fissured	Dilated with a gaping ileo-caecal valve
Cancer risk	Uncommon	High
Pseudodiverticulation	Common	Not seen
Inflammatory polyps	Rare	Seen in about 20%
Gallstones	Common especially after surgery	No significant increase

Perbedaan Chron's dan Ulcerative colitis



Radiologi Canggih

- **USG**
- **CT**
- **MRI** dan **MRCP** (Magnetic Resonance Cholangiopancreatography)
- **PTC** (Percutaneous Transhepatic Cholangiography)
- **ERCP** (Endoscopic Retrograde Cholangiopancreatography)
- **T-Tube Cholangiography**
- **Nuklir**

CT-scan pd Ileus

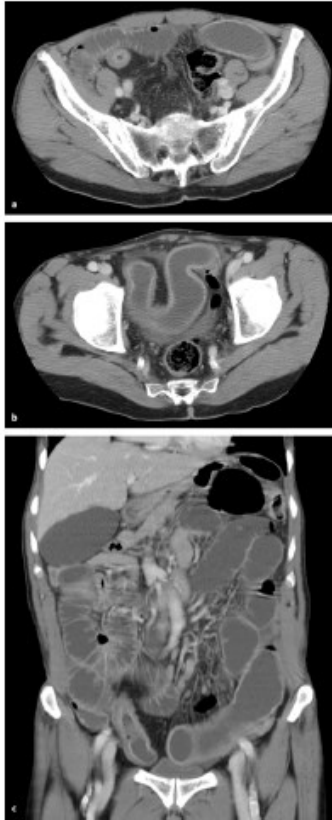


Fig. 4. Small bowel obstruction due to Crohn's disease. Axial slices (a, b) and coronal reformattimg (c) show a thickening of the wall of the distal ileum due to Crohn's disease responsible for a SBO. Note also a scleropermeatosis characteristic of the disease



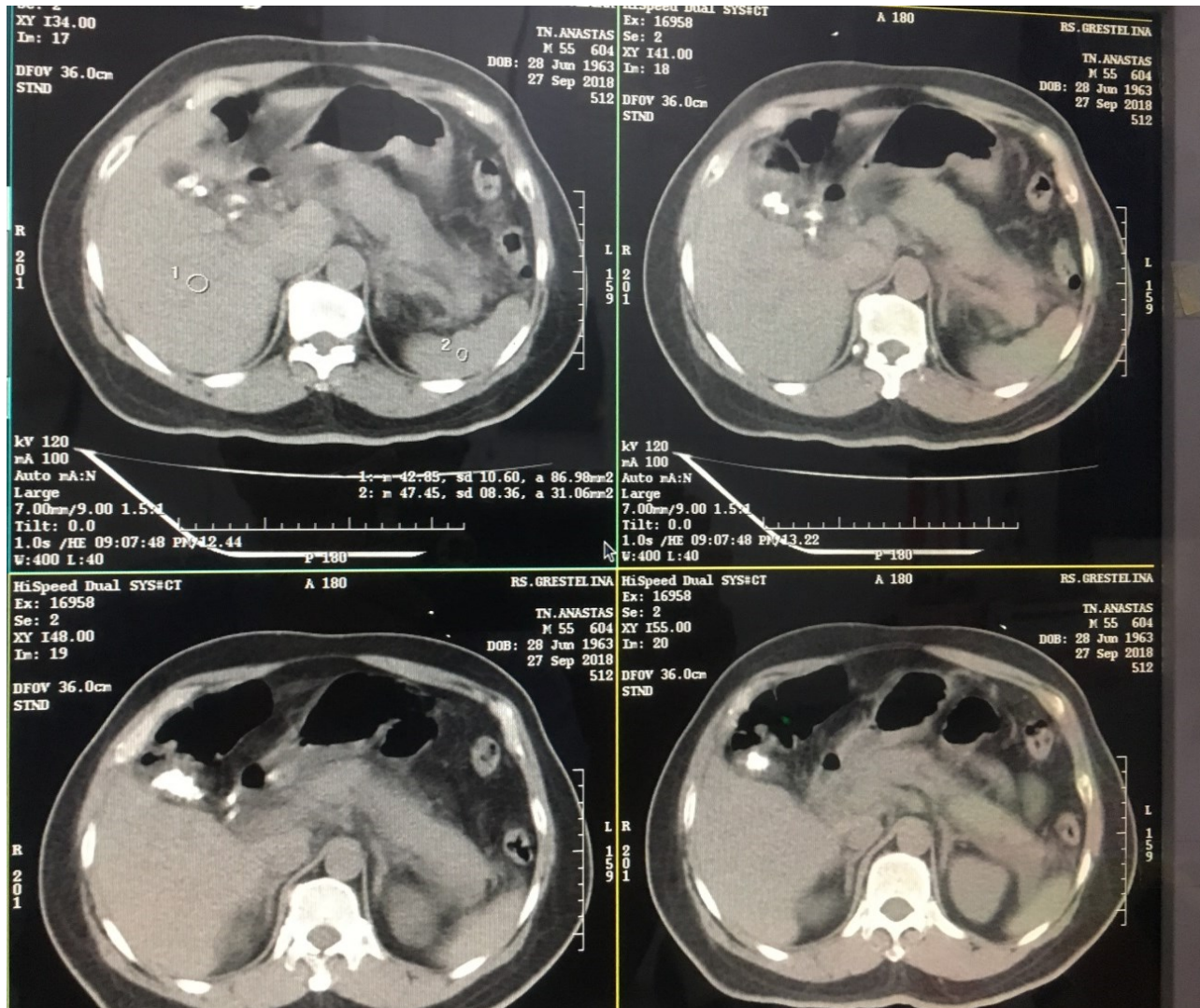
Fig. 5 a, b. Small bowel obstruction due to adhesive band. Oblique axial and coronal reformattimg clearly show the transition zone (arrow) between the dilated and collapsed bowel



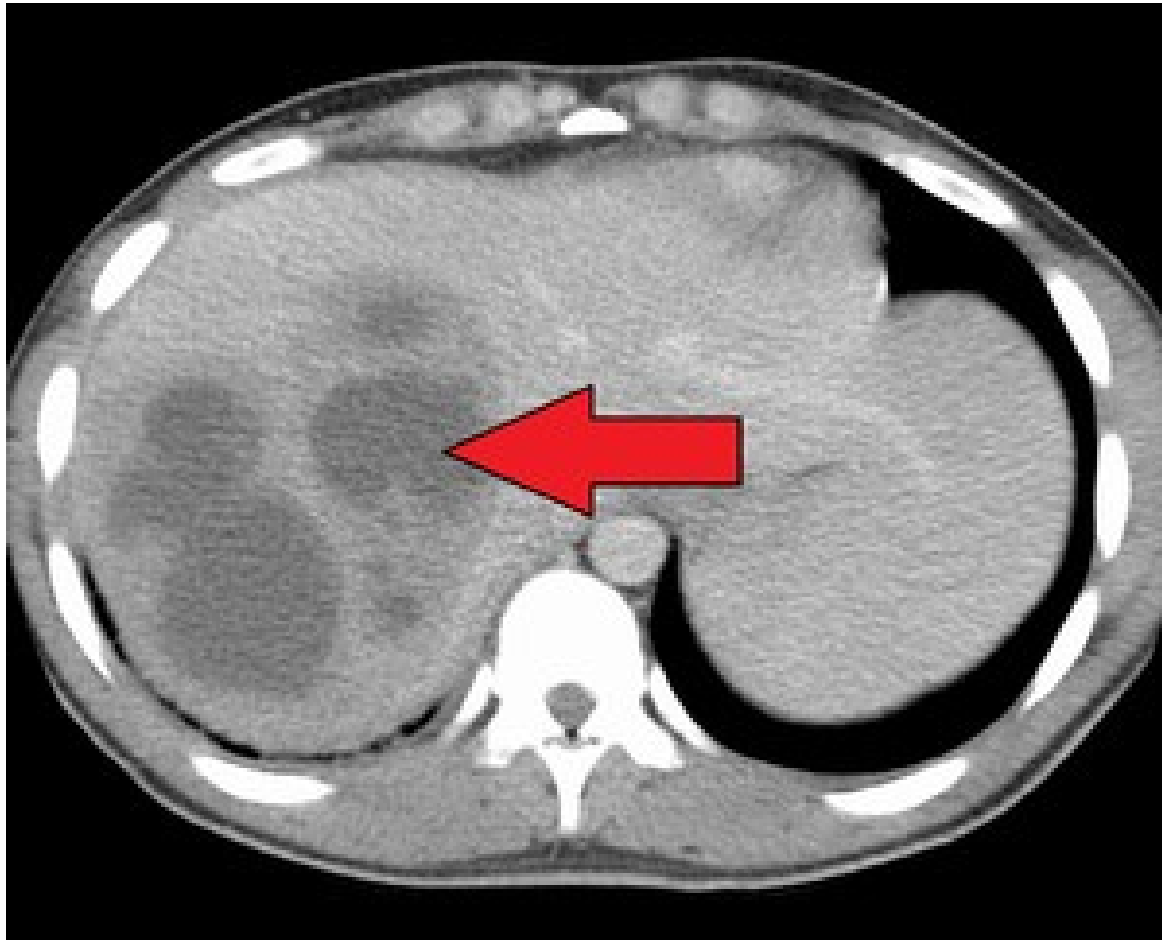
Nyeri perut atas

Pankreatitis akut

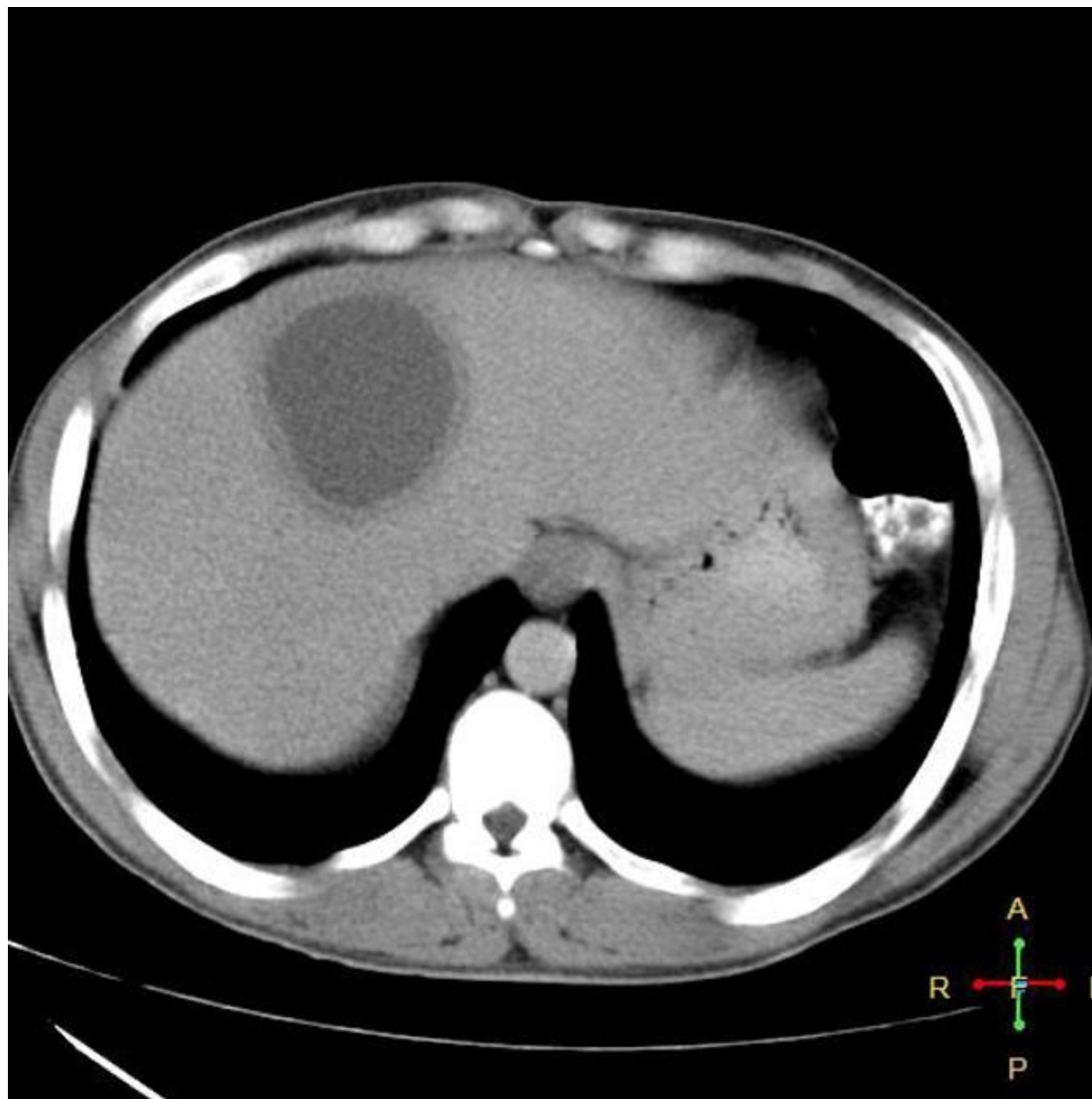
(cholecystitis cholelithiasis dan pankreatitis akut



Nyeri abdomen kanan atas dan leukositosis



Abses hepar



Sakit perut kanan atas

- * Gallbladder wall thickening > 3 mm

- * Sludge

Cholecystitis akut dgn batu
(ada hiperekhoik dgn
acoustic shadowing)

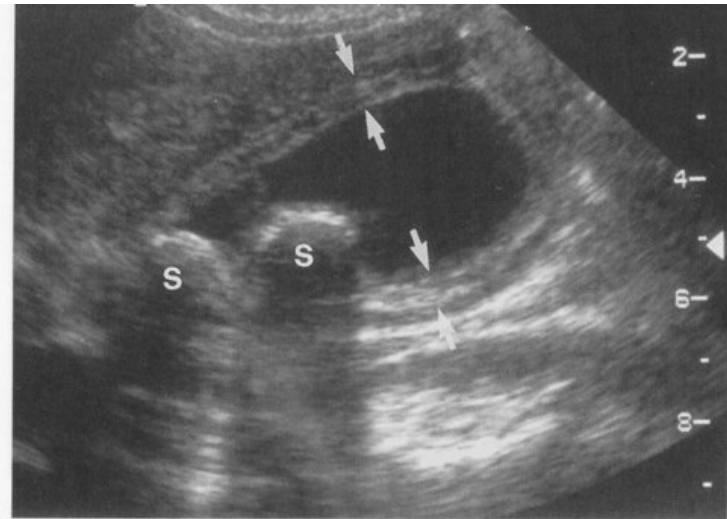


Figure 1-8 Acute cholecystitis with gallbladder wall thickening. Longitudinal view of the gallbladder demonstrates two stones (s) in the gallbladder. The gallbladder wall is diffusely thickened (**arrows**). This patient also demonstrated a positive sonographic Murphy sign and these findings are typical of acute cholecystitis. (Reproduced with permission from **Ultrasound: The Requisites**, editors, A.B. Kurtz and W.D. Middleton. Mosby Yearbook, St. Louis, 1996.)

- Soliter / multiple
- Echogenic/hyperechoic structure dengan *acoustic shadowing*

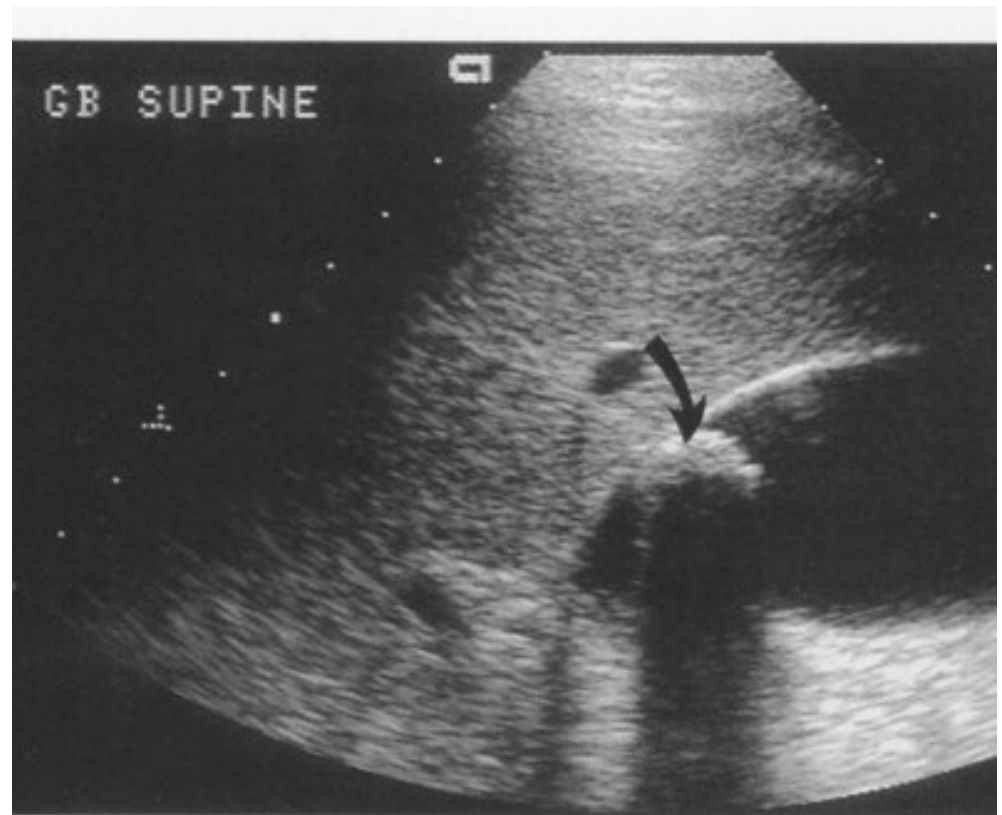
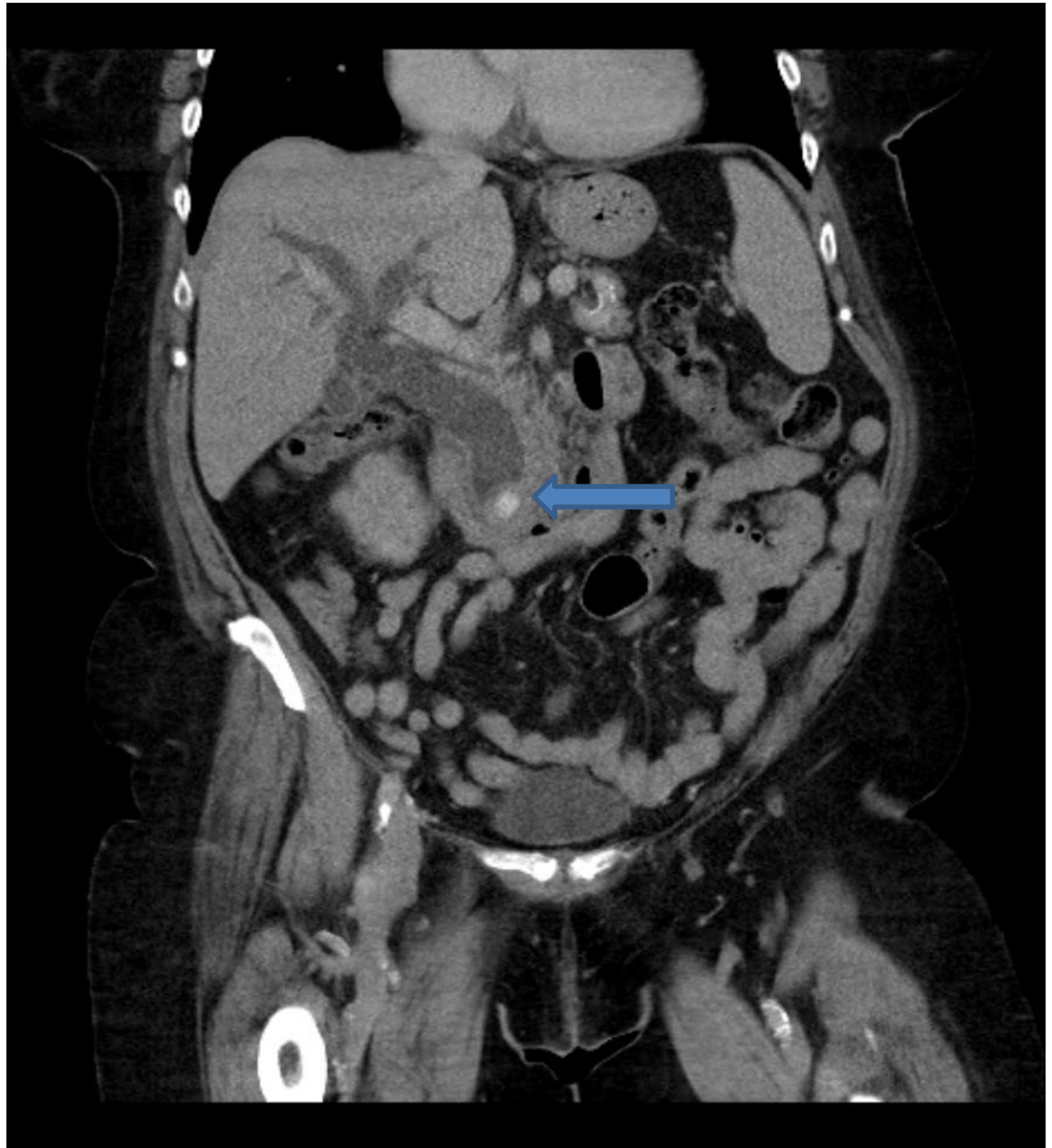


Figure 1-1 Typical gallstone. **A:** Longitudinal scan showing shadowing echogenic structure (arrow) near the neck of the gallbladder. **B:** Longitudinal scan with the patient in a left lateral

Ikterus,
intermittent

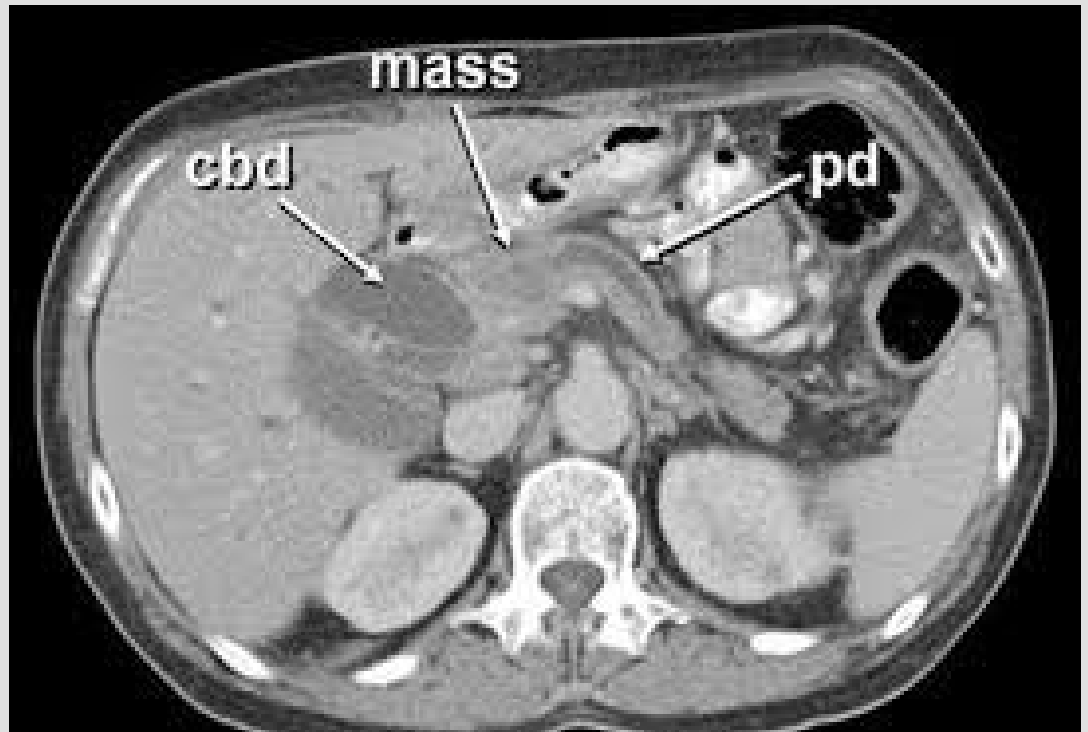
Obstruksi
CBD
oleh batu



Ikterus, dan
teraba
massa
epigastrium

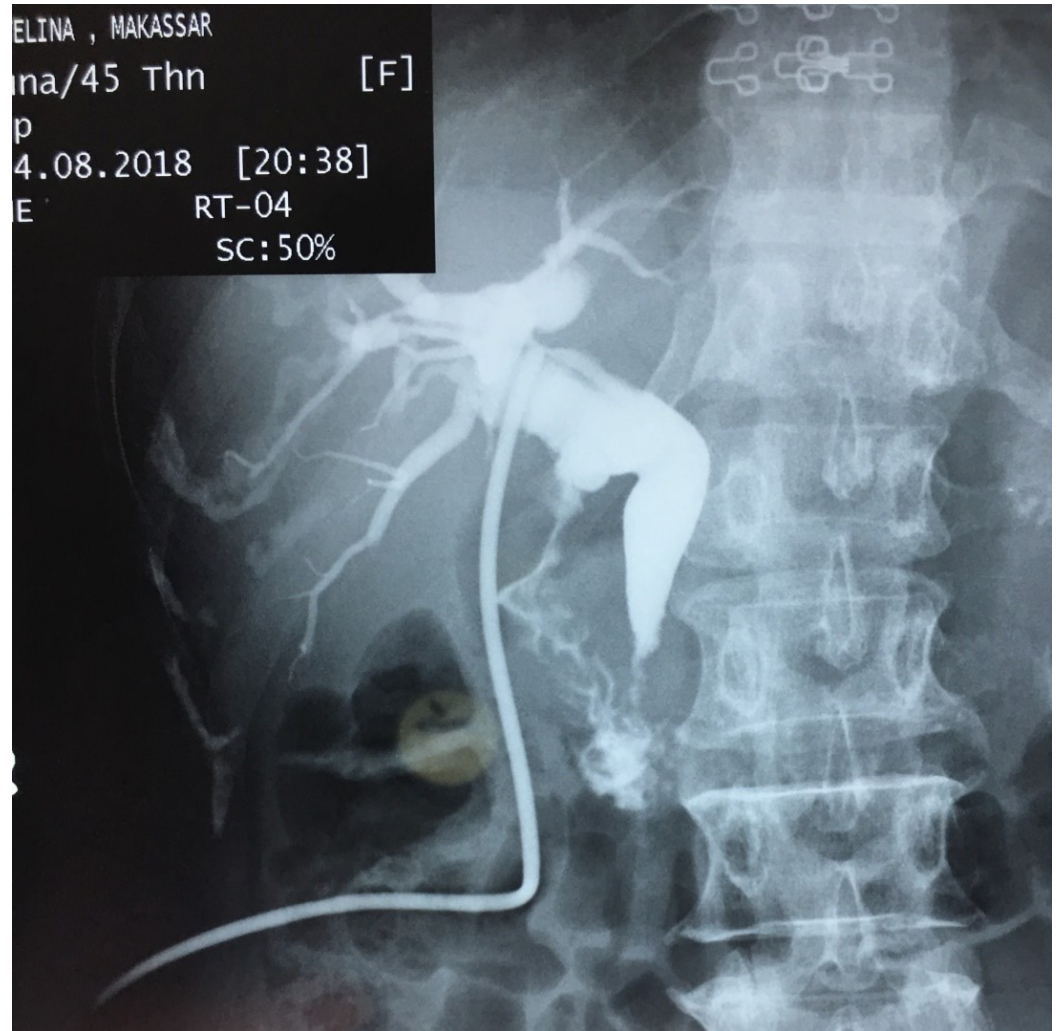
Obstruksi CBD
oleh

Tumor kaput
pankreas



T-Tube Cholangiography

- Post-operasi ikterus,
- Masih kuning

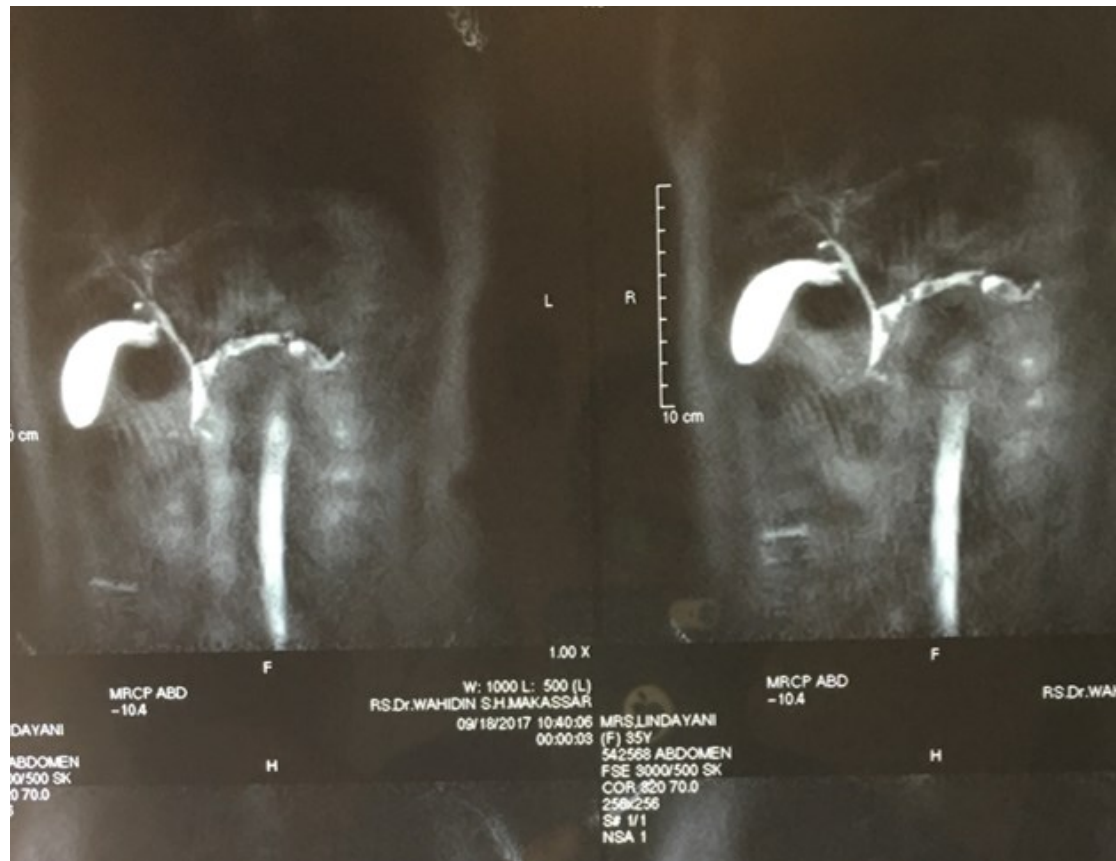


Perempuan 45 tahun dengan post eksplorasi bilier



T-Tube
cholangiography

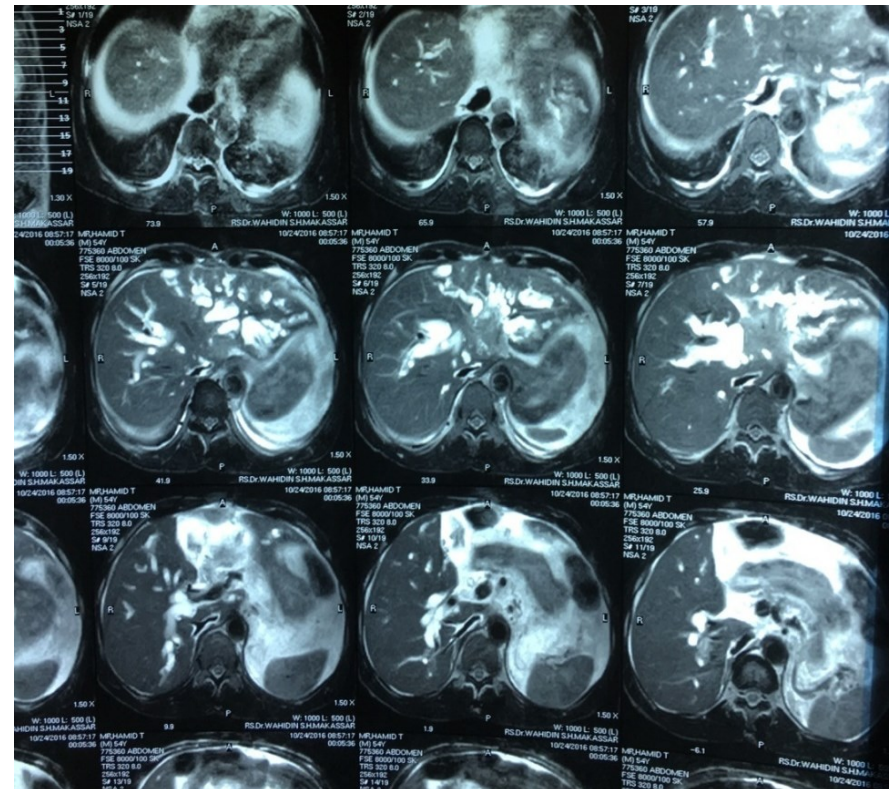
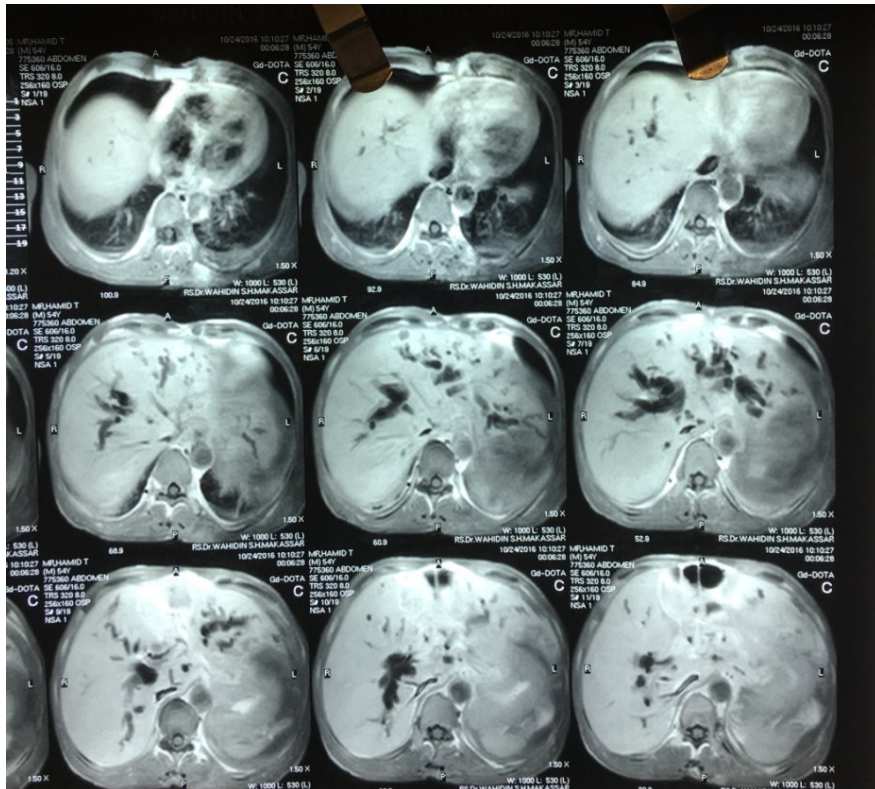
batu multiple pd
duktus pankreas

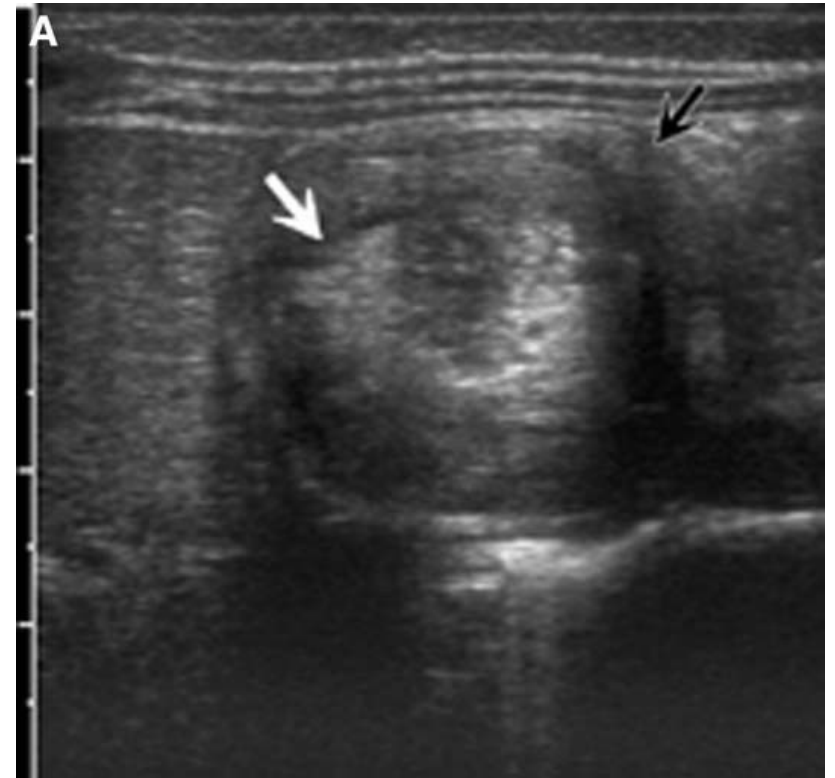
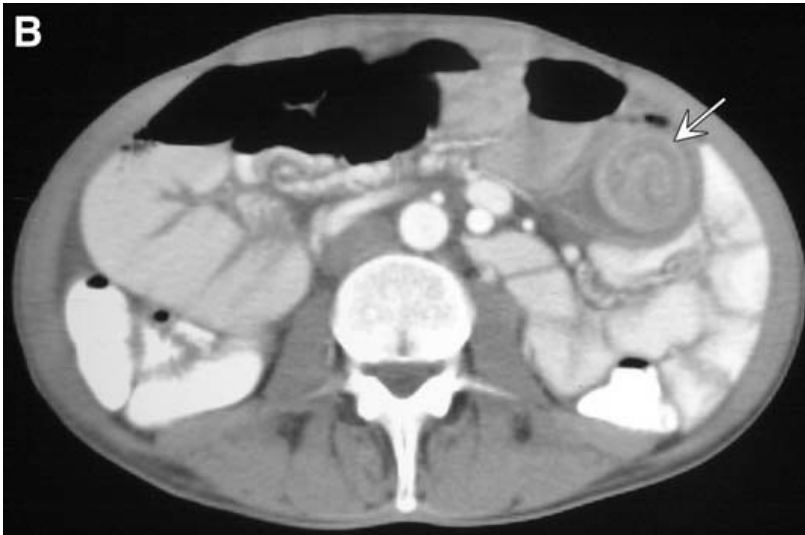


MRCP

MRI

- Laki2 50th dengan ikterus
- Obstruksi bilier oleh Tumor Cholangiocarcinoma

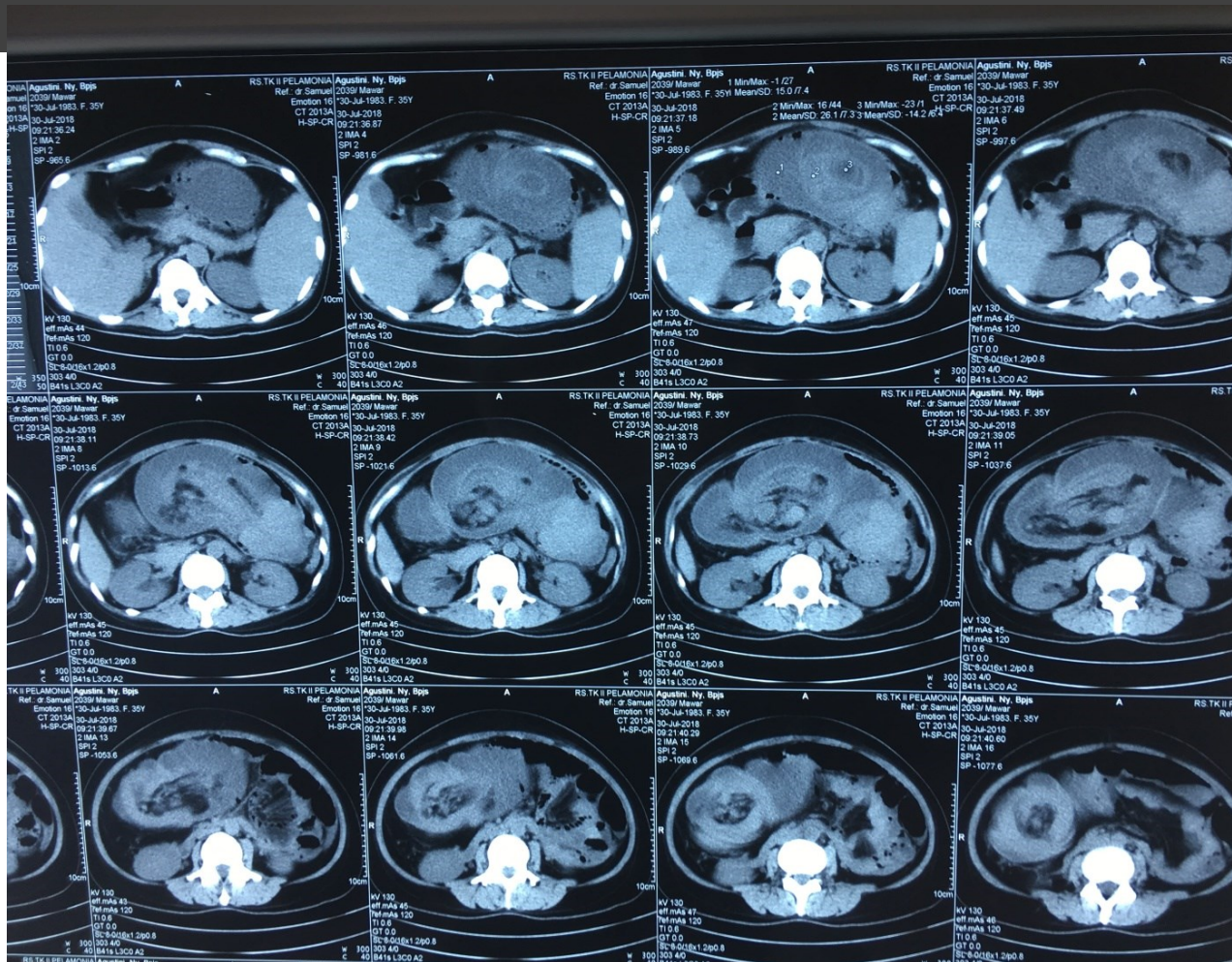


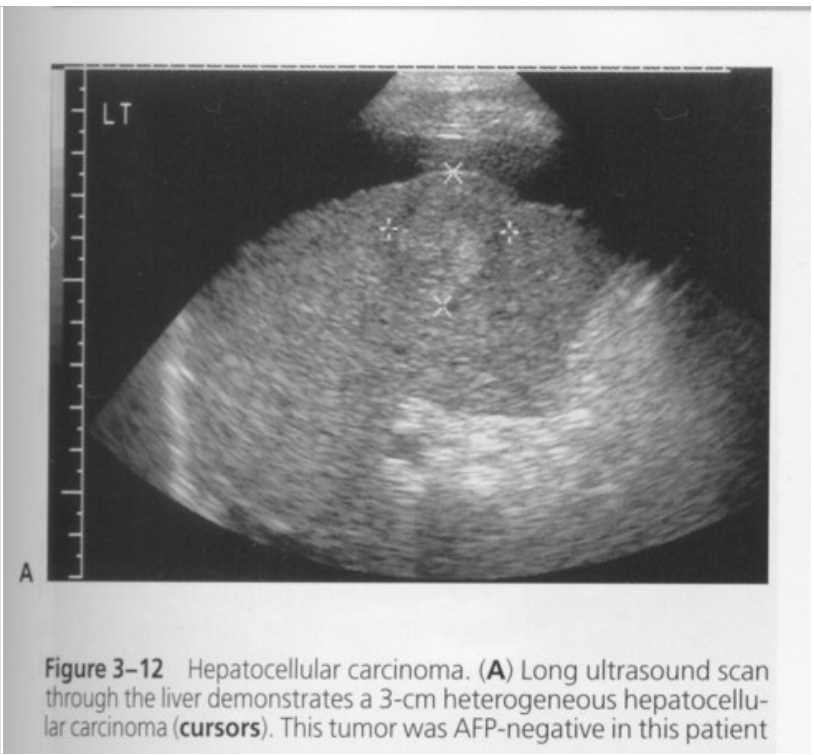
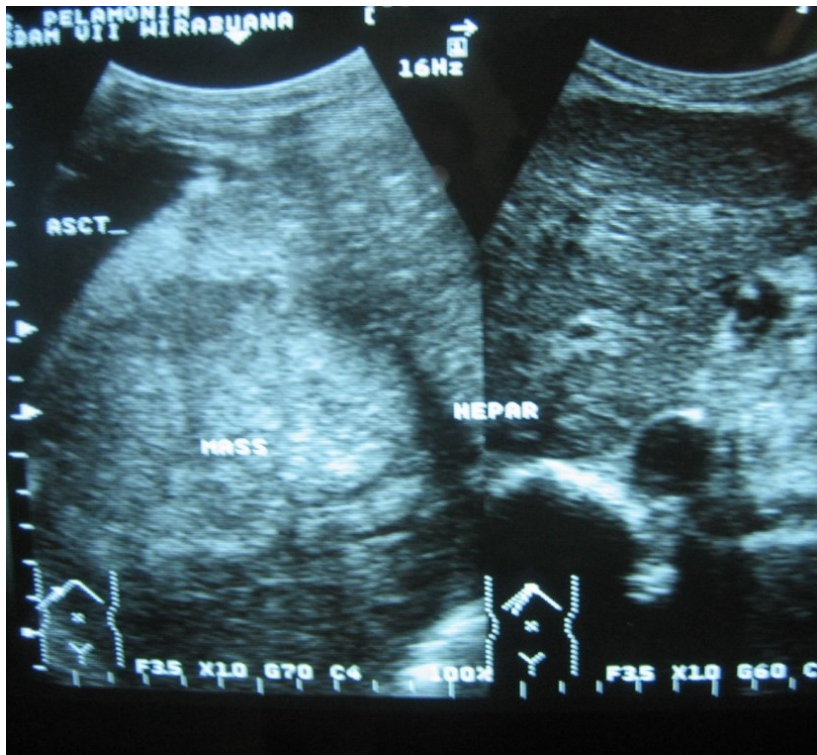


Sakit perut, dgn bab darah-lendir (INVAGINASI)

- Target sign, doughnut sign or bull's eye sign (transverse scan)
- pseudokidney sign (longitudinal scan)

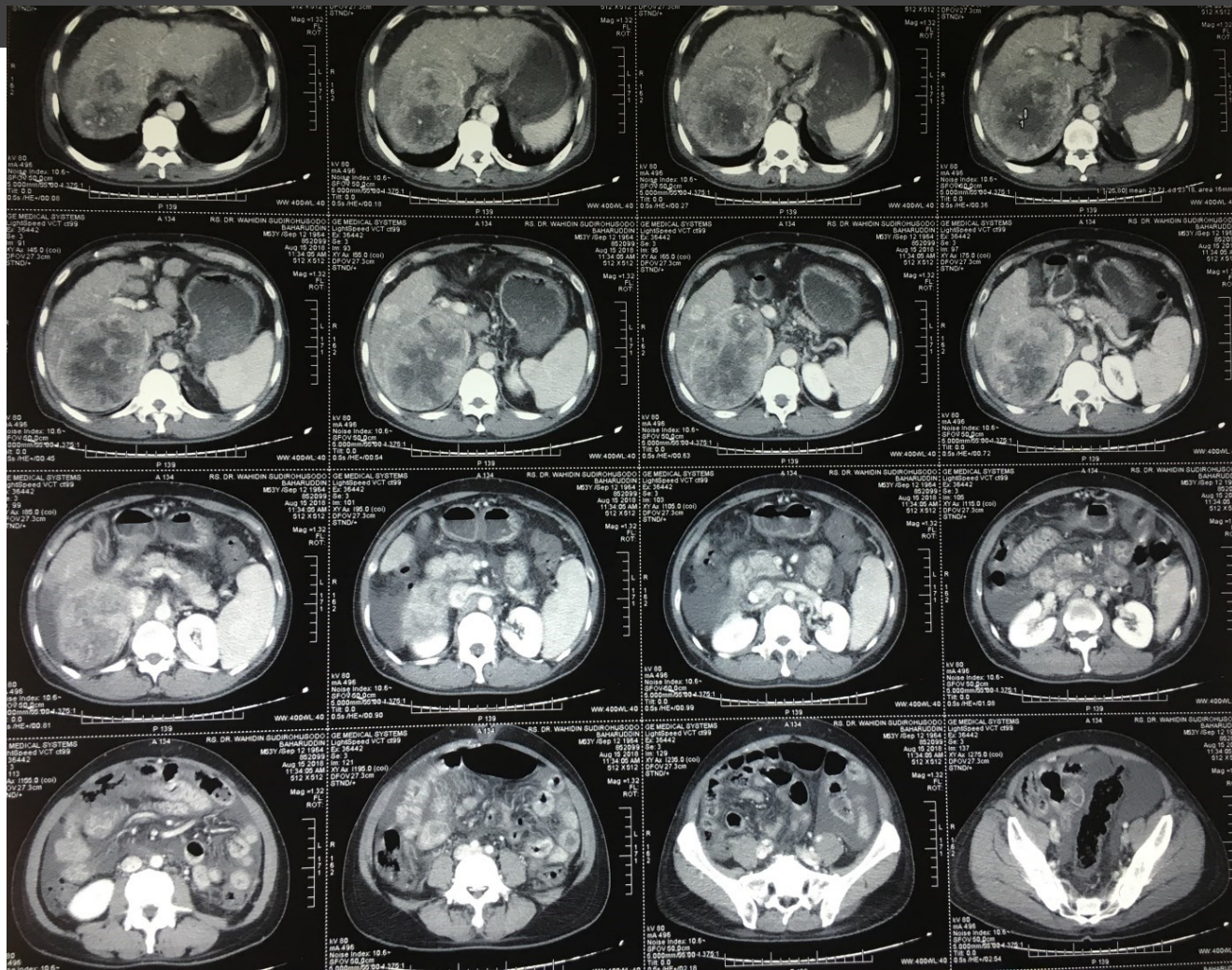
bab berdarah dan lendir (INVAGINASI)



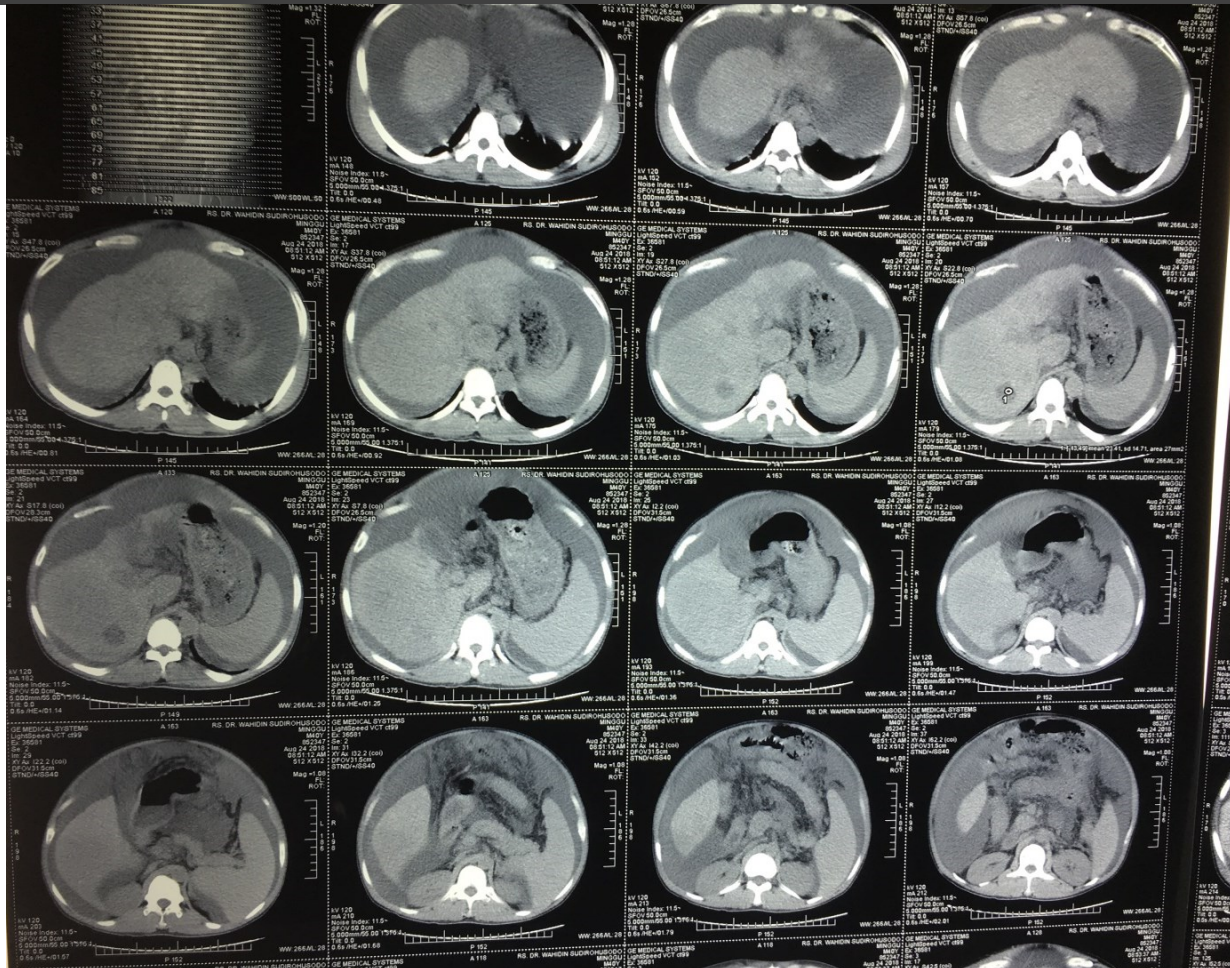


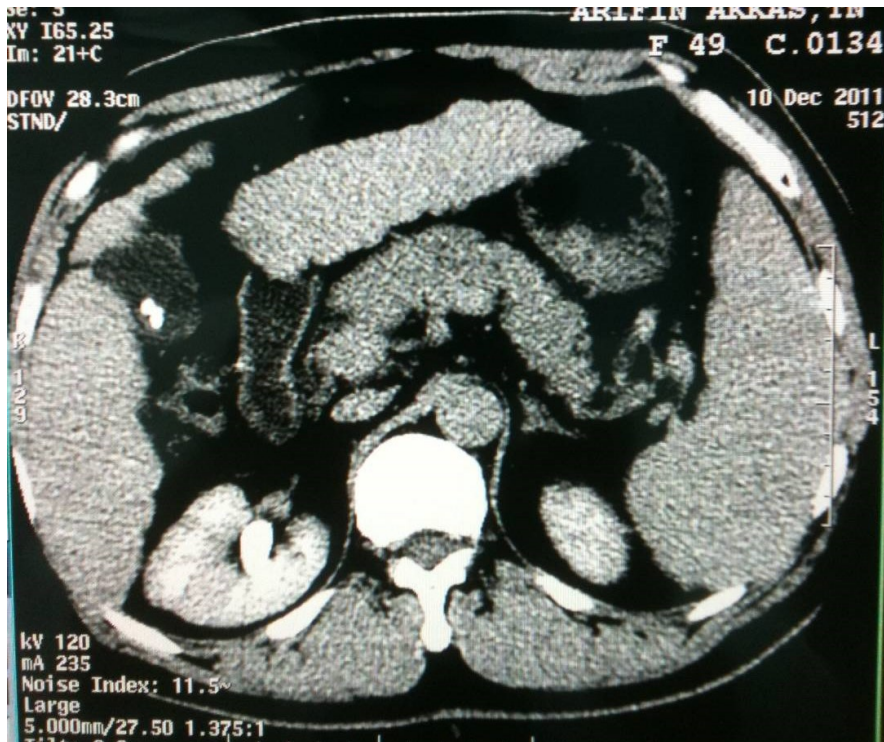
Massa hepar (HEPATOMA)

massa hipokhondrium kanan, suatu Hepatoma

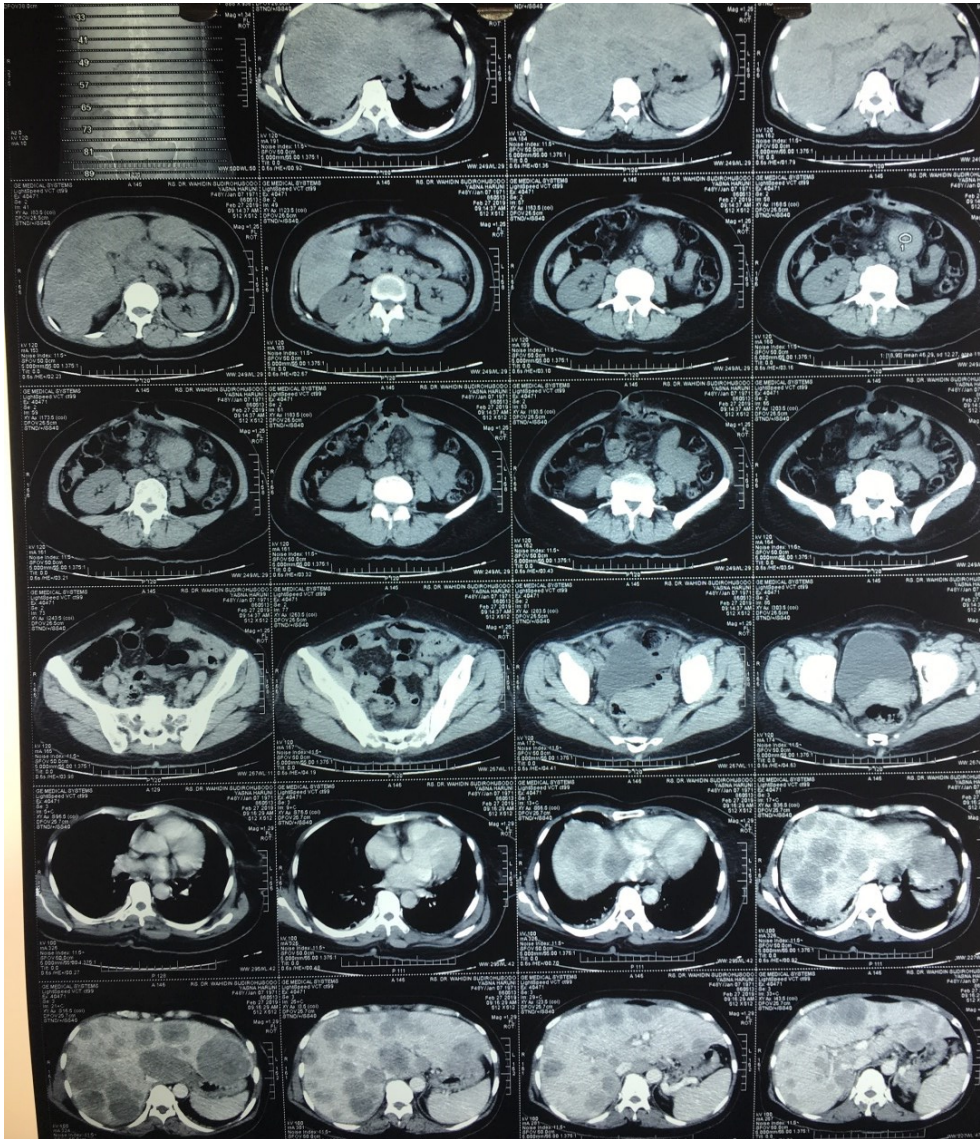


Sirosis hepatis dgn hepatoma, disertai ascites





Cirrhosis hepatis
Hepar tampak
atrofi,
permukaan
irreguler, lien
membesar



Metastasis
tampak nodul
multiple pd
hepar

KESIMPULAN

- Pemeriksaan radiologi abdomen (konvensional) terdiri dari foto **polos** (plain radiography) dan foto **kontras** (Barium atau Iodium)
- Pd kasus emergensi diperlukan foto 3 posisi : **supine, erect, dan Lateral Decubitis kiri**
- Pemeriksaan **Imejing Canggih** meliputi : **USG,CT Scan, MRI/MRCP, PTC, ERCP, NUKLIR.**

Terima Kasih

Selamat Belajar

Bayi setiap kali menyusui, muntah





VectorStock

vectorstock.com/22731878



Frimann-Dahl sign

